HHL Quantum Linear-Systems Algorithm

Assessment of the Harrow–Hassidim–Lloyd algorithm

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key**, section by section, with the topic tested by every question (page 2),
2. a **topic-wise map** grouping the questions by theme and a **quick-marking grid** (page 3), and
3. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 4).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count N_c = number of correct answers and N_w = number of questions answered wrongly (or with more than one option marked). Section marks = $3N_c - N_w$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy linear algebra, a full NumPy simulation of the notebook's HHL circuit, and CUDA-Q runs of the notebook) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- Linear system $Ax = b$, A a $d \times d$ matrix; $|b\rangle = b/\|b\|$, $|x\rangle = x/\|x\|$. For Hermitian A: $A = \sum_j \lambda_j |u_j\rangle\langle u_j|$, $\beta_j = \langle u_j|b\rangle$; $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ (for general matrices $\sigma_{\max}/\sigma_{\min}$); sparsity s = maximal number of non-zeros per row.
- Basis states are written with the **most significant bit first**; $\mathrm{QFT}\,|k\rangle = 2^{-n/2}\sum_y e^{2\pi iky/2^n}|y\rangle$; $R_Y(\theta) = e^{-i\theta Y/2}$.
- **Phase estimation** with t counting qubits on $U|u\rangle = e^{2\pi i\varphi}|u\rangle$: Hadamards, controlled U^{2^j}, inverse QFT; outcome $y \approx 2^t\varphi$.
- **HHL in the notebook** (`HHL.ipynb`): n_c clock qubits, $n_b = \log_2 d$ system qubits, 1 ancilla; $U = e^{iAt}$; clock value of λ: $k = \lambda t\, 2^{n_c}/(2\pi)$, signed (values $k \geq 2^{n_c-1}$ mean $k - 2^{n_c}$); $C = 2\pi/(2^{n_c}t)$; inversion $R_Y(2\arcsin(C/\lambda_k))$ on the ancilla; success = ancilla 1 and clock 0, with $P_{\text{success}} = C^2\|A^{-1}b\|^2/\|b\|^2$.
- **Default example:** $A = H\,\mathrm{diag}(1,2,3,4)H$ with $H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$, $b = (1,0,0,0)$, $n_c = 4$, $t = \pi/8$, $C = 1$, 7 qubits. A *previously recorded run* means the outputs stored in the notebook (CUDA-Q 0.16.0, CPU target `qpp-cpu`).
- Numerical answers are rounded.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Linear Algebra and Mathematical Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**B**	Solving with an eigenvector	11	**C**	Conjugate-gradient scaling
2	**D**	Eigenvalues and κ (2×2)	12	**D**	Amplitude encoding
3	**C**	Spectra of matrix classes	13	**B**	$\lvert b\rangle$ in the eigenbasis
4	**D**	Condition number	14	**A**	2×2 example: $x = A^{-1}b$
5	**D**	Inverse via eigen-decomposition	15	**C**	2×2 example: $\lvert x\rangle$
6	**A**	Hermitian embedding; singular values	16	**B**	e^{iAt} of a diagonal matrix
7	**A**	$\kappa \geq 1$	17	**B**	Unitarity of e^{iAt}
8	**D**	Hermitian vs. invertible; spectral theorem	18	**A**	Bounds on $\lVert A^{-1}b\rVert$
9	**C**	Sparsity	19	**B**	Singular matrices
10	**A**	Cost of Gaussian elimination	20	**C**	Scaling and κ

Section B — Quantum Computing Foundations

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**C**	Exact phase estimation	31	**A**	Post-selection
22	**D**	Clock value of an eigenvalue	32	**C**	System–clock entanglement
23	**D**	Stages of phase estimation	33	**C**	Memory of the HHL state
24	**A**	Phase kickback	34	**D**	Cost of state preparation
25	**B**	Inverse QFT of QFT $\lvert 1\rangle$	35	**B**	Measuring the clock too early
26	**D**	QFT gate counts in HHL	36	**A**	Swap test
27	**B**	$R_Y(2 \arcsin c)$	37	**B**	QPE with $\varphi = 1/4$
28	**B**	Polarity X gates	38	**D**	QPE leakage, $\varphi = 1/3$
29	**A**	Controlled powers U^{2^j}	39	**C**	QFT unitarity and cost
30	**C**	Two's complement clock values	40	**A**	Shift $A \to A + I$

Section C — HHL Algorithm Fundamentals

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**D**	State after inversion	51	**B**	Negative clock values
42	**B**	Qubit count	52	**C**	Exact vs. rounded clock values
43	**A**	Role of the constant C	53	**D**	2×2 example: t, C
44	**D**	Rotation angle	54	**C**	Post-selection condition
45	**C**	Success probability (default)	55	**A**	Shot statistics of P_{success}
46	**A**	Register transformations	56	**B**	What sampling reveals
47	**D**	Automatic evolution time	57	**B**	Multi-qubit gate count
48	**D**	Why uncompute	58	**C**	HHL $\leftrightarrow$ code
49	**A**	Norm recovery	59	**C**	Passage: state after QPE
50	**B**	Number of inversion rotations	60	**A**	Passage: P_{success} and $\lvert x\rangle$

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Section D — Advanced HHL and Implementation

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**D**	HHL complexity	71	**C**	Aliasing
62	**A**	Amplitude amplification	72	**C**	Quantum linear-system algorithms
63	**C**	Read-out cost	73	**A**	Density matrix and global phase
64	**D**	Clock size for κ	74	**A**	Custom gates; hardware refused
65	**A**	Exponential rotation count	75	**B**	Too few clock qubits
66	**B**	Embedding in the notebook	76	**B**	Memory with 8 clock qubits
67	**A**	Padding and κ	77	**C**	How e^{iAt} is implemented
68	**D**	Dequantisation	78	**D**	Useful outputs
69	**D**	Conditional speed-up	79	**B**	Recorded run: shot noise
70	**B**	Worst-case success probability	80	**C**	Recorded run: $\|x\|$ from P_{success}

Topic-wise map of the questions

Every question appears under at least one theme; several appear under two.

Theme	Question numbers
Linear systems, inverses and eigen-decomposition	1, 5, 13, 14, 15, 18, 41
Matrix properties: Hermitian, unitary, singular, embedding	3, 6, 8, 16, 17, 19, 66
Condition number and sparsity	2, 4, 7, 9, 20, 64, 67, 70
Classical solvers and complexity	10, 11, 61, 68, 69
State preparation and read-out	12, 34, 36, 56, 63, 78
Phase estimation and the QFT	21, 22, 23, 24, 25, 26, 29, 37, 38, 39, 40
Clock values, two's complement and aliasing	30, 47, 51, 52, 53, 71, 75
Eigenvalue inversion and the constant C	27, 28, 43, 44, 50, 65
Uncomputation, entanglement and post-selection	31, 32, 35, 46, 48, 54, 59, 60
Success probability and amplitude amplification	45, 55, 62, 70, 80
The notebook's circuit, code and memory	33, 42, 49, 57, 58, 73, 74, 76, 77
Newer algorithms and recorded results	72, 79, 80

Quick-marking grid

1 **B**	2 **D**	3 **C**	4 **D**	5 **D**	6 **A**	7 **A**	8 **D**	9 **C**	10 **A**
11 **C**	12 **D**	13 **B**	14 **A**	15 **C**	16 **B**	17 **B**	18 **A**	19 **B**	20 **C**
21 **C**	22 **D**	23 **D**	24 **A**	25 **B**	26 **D**	27 **B**	28 **B**	29 **A**	30 **C**
31 **A**	32 **C**	33 **C**	34 **D**	35 **B**	36 **A**	37 **B**	38 **D**	39 **C**	40 **A**
41 **D**	42 **B**	43 **A**	44 **D**	45 **C**	46 **A**	47 **D**	48 **D**	49 **A**	50 **B**
51 **B**	52 **C**	53 **D**	54 **C**	55 **A**	56 **B**	57 **B**	58 **C**	59 **C**	60 **A**
61 **D**	62 **A**	63 **C**	64 **D**	65 **A**	66 **B**	67 **A**	68 **D**	69 **D**	70 **B**
71 **C**	72 **C**	73 **A**	74 **A**	75 **B**	76 **B**	77 **C**	78 **D**	79 **B**	80 **C**

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Answer Bank — questions with detailed solutions

SECTION A · Linear Algebra and Mathematical Foundations Q.1–20

Q.1 SINGLE-CORRECT MCQ · Solving with an eigenvector

The solution of $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is

(A) $x = (0.5,\ 0.5)^{\mathsf{T}}$
(B) ✓ $x = (0.5,\ -0.5)^{\mathsf{T}}$
(C) $x = (0.25,\ -0.25)^{\mathsf{T}}$
(D) $x = (1,\ -1)^{\mathsf{T}}$

Answer: (B) Detailed solution

Correct answer. (B) $x = (0.5,\ -0.5)^{\mathsf{T}}$
Concepts required. Eigenvectors are scaled by A; inverting A divides each eigencomponent by its eigenvalue.
Relevant formulae. $Av = \lambda v \Rightarrow A^{-1}v = v/\lambda$; $A^{-1}b = \sum_j \frac{\langle u_j|b\rangle}{\lambda_j} u_j$.
Step-by-step explanation. $b = (1,-1)$ is an eigenvector of A: $A(1,-1)^{\mathsf{T}} = (3-1, 1-3) = (2,-2) = 2(1,-1)$, eigenvalue $\lambda = 2$. For an eigenvector, $A^{-1}b = b/\lambda = (0.5,-0.5)$. Check: $A(0.5,-0.5) = (1.5-0.5, 0.5-1.5) = (1,-1)$ ✓. This is the idea of HHL in one line: in the eigenbasis, inverting A means dividing each component by its eigenvalue.
Why the other options are incorrect.
(A) $(0.5, 0.5)$ has the wrong sign in the second component: $A(0.5,0.5) = (2,2) \neq b$.
(C) $(0.25, -0.25)$ divides by 4, the *other* eigenvalue (eigenvector $(1,1)$).
(D) $(1,-1)$ is b itself; it would solve the system only if $\lambda = 1$.
Key takeaway. In the eigenbasis, A^{-1} is division by λ — exactly what HHL implements.

Q.2 SINGLE-CORRECT MCQ · Eigenvalues and κ (2×2)

The eigenvalues and the condition number $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ of $A = \begin{pmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix}$ are

(A) $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = \frac{2}{3}$
(B) $\frac{8}{9}$ and 1; $\kappa = \frac{9}{8}$
(C) 1 and $-\frac{1}{3}$; $\kappa = 3$
(D) ✓ $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = 2$

Answer: (D) Detailed solution

Correct answer. (D) $\frac{2}{3}$ and $\frac{4}{3}$; $\kappa = 2$
Concepts required. Eigenvalues of a symmetric 2×2 matrix with equal diagonal; condition number.
Relevant formulae. $\left(\begin{smallmatrix} a & c \\ c & a \end{smallmatrix}\right)$: $\lambda = a \pm c$; $\kappa = |\lambda|_{\max}/|\lambda|_{\min} \geq 1$.
Step-by-step explanation. For $\left(\begin{smallmatrix} a & c \\ c & a \end{smallmatrix}\right)$ the eigenvalues are $a \pm c$ with eigenvectors $(1, \pm 1)/\sqrt{2}$: here $1 - \frac{1}{3} = \frac{2}{3}$ and $1 + \frac{1}{3} = \frac{4}{3}$. $\kappa = \frac{4/3}{2/3} = 2$ (always ≥ 1). Check: trace $= 2 = \frac{2}{3} + \frac{4}{3}$, determinant $= 1 - \frac{1}{9} = \frac{8}{9} = \frac{2}{3} \cdot \frac{4}{3}$. $\frac{8}{9}$ is the determinant, not an eigenvalue; 1 and $-\frac{1}{3}$ are matrix entries.
Why the other options are incorrect.
(A) $\kappa = \frac{2}{3}$ divides the small by the large eigenvalue; κ is always ≥ 1.
(B) $\frac{8}{9}$ is the determinant (product of the eigenvalues).
(C) 1 and $-\frac{1}{3}$ are entries of A, not eigenvalues.
Key takeaway. Check eigenvalues with trace (sum) and determinant (product).

Q.3 MATCHING LISTS · Spectra of matrix classes

Match each class of square matrix (List-I) with the property of its eigenvalues (List-II).

List-I		List-II	
(P)	Hermitian ($A = A^\dagger$)	**(1)**	at least one eigenvalue is 0
(Q)	unitary ($U^\dagger U = I$)	**(2)**	all eigenvalues are real
(R)	Hermitian positive definite	**(3)**	all eigenvalues lie on the unit circle, $\lvert\lambda\rvert = 1$
(S)	singular	**(4)**	all eigenvalues are real and strictly positive

(A) P→2, Q→4, R→3, S→1
(B) P→2, Q→3, R→1, S→4
(C) ✓ P→2, Q→3, R→4, S→1
(D) P→3, Q→2, R→4, S→1

Answer: (C) Detailed solution

Correct answer. **(C)** P→2, Q→3, R→4, S→1
Concepts required. Spectral properties of matrix classes.
Relevant formulae. Hermitian: $\lambda \in \mathbb{R}$; unitary: $|\lambda| = 1$; positive definite: $\lambda > 0$; singular: $\det = \prod \lambda = 0$.
Step-by-step explanation.

- P: $Av = \lambda v$, $\|v\| = 1$: $\lambda = v^\dagger A v = (v^\dagger A v)^* = \lambda^*$, real (2).
- Q: $\|Uv\| = \|v\|$ forces $|\lambda| = 1$ (3) — e.g. $U = e^{iAt}$ has eigenvalues $e^{i\lambda t}$.
- R: positive definite: $v^\dagger A v > 0$ for all $v \neq 0$, so every eigenvalue is real and positive (4).
- S: singular $\iff$ $\det A = \prod \lambda_i = 0$ $\iff$ some eigenvalue is 0 (1); then A^{-1} does not exist.

HHL needs a Hermitian (or embedded) invertible matrix; the phase estimation step works with the unitary e^{iAt}.
Why the other options are incorrect.
(A) Q→4 claims unitary eigenvalues are positive reals; $e^{i\lambda t}$ shows they are phases.
(B) R→1, S→4 calls a positive-definite matrix singular; positive-definite matrices are invertible.
(D) P→3, Q→2 swaps Hermitian and unitary: unitary eigenvalues are complex phases, Hermitian ones are real.
Key takeaway. HHL turns real eigenvalues (Hermitian A) into phases (unitary e^{iAt}).

Q.4 MATCHING LISTS · Condition number

Match each matrix (List-I) with its condition number $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$ (List-II).

List-I		List-II	
(P)	$\mathrm{diag}(0.5,\ 2,\ 8)$	**(1)**	4
(Q)	$\begin{pmatrix} 1 & -1/3 \\ -1/3 & 1 \end{pmatrix}$	**(2)**	1
(R)	the notebook's default A (eigenvalues $1, 2, 3, 4$)	**(3)**	16
(S)	$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$	**(4)**	2

(A) P→1, Q→4, R→3, S→2
(B) P→3, Q→4, R→2, S→1
(C) P→3, Q→2, R→1, S→4
(D) ✓ P→3, Q→4, R→1, S→2

Answer: (D) **Detailed solution**

Correct answer. **(D)** P→3, Q→4, R→1, S→2
Concepts required. Condition number of Hermitian matrices.
Relevant formulae. $\kappa = |\lambda|_{\max}/|\lambda|_{\min} \geq 1$; $\kappa(cI) = 1$.
Step-by-step explanation.

- P: $8/0.5 = 16$ (3); the middle eigenvalue does not matter.
- Q: eigenvalues $\frac{2}{3}, \frac{4}{3}$: $\kappa = 2$ (4).
- R: $4/1 = 4$ (1), the value the notebook prints.
- S: a multiple of the identity has $\kappa = 1$ (2) — the best possible.

HHL's cost and failure probability grow with κ (success probability $\geq 1/\kappa^2$).
Why the other options are incorrect.
(A) P→1, R→3 swaps the diagonal matrix (16) and the default (4).
(B) R→2, S→1 gives the default $\kappa = 1$ and $2I$ $\kappa = 4$.
(C) Q→2, S→4 gives the 2×2 example $\kappa = 1$; its eigenvalues differ ($\frac{2}{3}$, $\frac{4}{3}$).
Key takeaway. Only the extreme eigenvalues matter; multiples of I are perfectly conditioned.

Q.5 SINGLE-CORRECT MCQ · Inverse via eigen-decomposition

The notebook's default matrix is $A = H\ \mathrm{diag}(1,2,3,4)\ H$ with the symmetric orthogonal matrix $H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ ($H^2 = I$). For $b = (1,0,0,0)^{\mathsf{T}}$, the first component x_0 of $x = A^{-1}b$ is

(A) $\frac{5}{8}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{25}{12} \approx 2.0833$ **(D)** ✓ $\frac{25}{48} \approx 0.5208$

Answer: (D) **Detailed solution**

Correct answer. **(D)** $\frac{25}{48} \approx 0.5208$
Concepts required. Inverse through the eigen-decomposition.
Relevant formulae. $A = H\Lambda H \Rightarrow A^{-1} = H\Lambda^{-1}H$; $(A^{-1})_{00} = \sum_j H_{0j}^2/\lambda_j$.
Step-by-step explanation. $A^{-1} = H\ \mathrm{diag}(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4})\ H$ (invert the eigenvalues, keep the eigenvectors). Then $x_0 = (A^{-1})_{00} = \sum_j H_{0j}^2\, \lambda_j^{-1} = \frac{1}{4}\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) = \frac{1}{4} \cdot \frac{25}{12} = \frac{25}{48} = 0.520833$, since every $H_{0j} = \frac{1}{2}$. This matches `np.linalg.solve` and the notebook: $x = (0.520833, 0.145833, 0.229167, 0.104167)$. $\frac{25}{12}$ forgets the factor $H_{0j}^2 = \frac{1}{4}$.
Why the other options are incorrect.
(A) $\frac{5}{8}$ adds the weights incorrectly ($\frac{1}{4}(1 + \frac{1}{2} + 1)$).
(B) $\frac{1}{4}$ uses only the largest eigenvalue's contribution once.
(C) $\frac{25}{12}$ forgets the weights $H_{0j}^2 = \frac{1}{4}$.
Key takeaway. $x_0 = \frac{1}{4}(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) = \frac{25}{48}$, matching the notebook.

Q.6 SINGLE-CORRECT MCQ · Hermitian embedding; singular values

HHL needs a Hermitian matrix, so the notebook embeds a non-Hermitian A as $\tilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}$.

For $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ the condition number of $\tilde{A}$ is

(A) ✓ $3 + 2\sqrt{2} \approx 5.83$ **(B)** $\sqrt{2} + 1 \approx 2.41$

(C) 1, because both eigenvalues of A are 1 **(D)** 2

Answer: (A) Detailed solution

Correct answer. **(A)** $3 + 2\sqrt{2} \approx 5.83$

Concepts required. Hermitian embedding; singular values.

Relevant formulae. eigenvalues of $\left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)$ are $\pm\sigma_i(A)$; $\sigma_i = \sqrt{\lambda_i(A^\dagger A)}$.

Step-by-step explanation. The eigenvalues of $\tilde{A}$ are $\pm\sigma_i$, the singular values of A. $A^\mathsf{T}A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 5 \end{smallmatrix}\right)$ has eigenvalues $3 \pm 2\sqrt{2}$, so $\sigma = \sqrt{3 \pm 2\sqrt{2}} = \sqrt{2} \pm 1$, i.e. 2.414 and 0.414. Hence $\tilde{A}$ has eigenvalues $\pm 2.414, \pm 0.414$ and $\kappa = \frac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2}+1)^2 = 3 + 2\sqrt{2} \approx 5.83$. The eigenvalues of a non-normal matrix (here both 1) say nothing about its conditioning; singular values do.

Why the other options are incorrect.

(B) $\sqrt{2} + 1$ is $\sigma_{\max}$ alone, not the ratio.

(C) Equal eigenvalues of a non-normal matrix say nothing about conditioning; singular values do.

(D) 2 has no derivation from the singular values $\sqrt{2} \pm 1$.

Key takeaway. Embedding preserves the singular-value condition number; non-normal matrices can be badly conditioned.

Q.7 ASSERTION–REASON · $\kappa \geq 1$

Assertion (A): Every invertible matrix has condition number $\kappa(A) = \|A\|\,\|A^{-1}\| \geq 1$.

Reason (R): In the spectral norm $\kappa(A) = \sigma_{\max}/\sigma_{\min}$, the ratio of the largest to the smallest singular value.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

Concepts required. Condition number via singular values.

Relevant formulae. $\kappa = \|A\|_2\|A^{-1}\|_2 = \sigma_{\max}/\sigma_{\min} \geq 1$.

Step-by-step explanation. **R is true:** $\|A\|_2 = \sigma_{\max}$ and $\|A^{-1}\|_2 = 1/\sigma_{\min}$. **A follows from R:** a ratio of the largest to the smallest of a set of positive numbers is at least 1 (equality for multiples of unitary matrices). For Hermitian A the singular values are $|\lambda_i|$, giving $\kappa = |\lambda|_{\max}/|\lambda|_{\min}$, the formula the notebook prints. Answer: (A).

Why the other options are incorrect.

(B) needs A and R both true but R not explaining A; R is precisely the reason.

(C) needs A true and R false; R is true.

(D) needs A false and R true; A is true.

Key takeaway. $\kappa \geq 1$, with equality only for multiples of unitary matrices.

Q.8 STATEMENT I / II · Hermitian vs. invertible; spectral theorem

Statement I: Every invertible matrix is Hermitian.

Statement II: Every real symmetric matrix can be written as $A = Q\Lambda Q^{\mathsf{T}}$ with an orthogonal Q and a real diagonal Λ.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Detailed solution

Correct answer. (D) Statement I is false, but Statement II is true
Concepts required. Hermiticity vs. invertibility; spectral theorem.
Relevant formulae. $A = Q\Lambda Q^{\mathsf{T}}$ for real symmetric A.
Step-by-step explanation. **I is false:** $\left(\begin{smallmatrix}1 & 2\\ 0 & 1\end{smallmatrix}\right)$ is invertible (det $= 1$) but not symmetric — the notebook embeds such matrices. **II is true:** the spectral theorem for real symmetric matrices; the notebook's default $A = H\,\mathrm{diag}(1,2,3,4)H$ is exactly of this form with $Q = H$. Answer: (D).
Why the other options are incorrect.
(A) needs both statements true; I is false.
(B) needs both statements false; II is true.
(C) needs I true and II false; I is false and II is true.
Key takeaway. Invertible ≠ Hermitian; HHL embeds non-Hermitian matrices.

Q.9 SINGLE-CORRECT MCQ · Sparsity

The sparsity s of a matrix is the largest number of non-zero entries in any row. For the notebook's default 4×4 matrix $A = \begin{pmatrix} 2.5 & -0.5 & -1 & 0 \\ -0.5 & 2.5 & 0 & -1 \\ -1 & 0 & 2.5 & -0.5 \\ 0 & -1 & -0.5 & 2.5 \end{pmatrix}$ and for a tridiagonal 1000×1000 matrix, s is

(A) 3 and 1000 **(B)** 12 and 2998 **(C)** ✓ 3 and 3 **(D)** 4 and 3

Answer: (C) Detailed solution

Correct answer. (C) 3 and 3
Concepts required. Sparsity as non-zeros per row.
Relevant formulae. $s = \max_i \#\{j : A_{ij} \neq 0\}$.
Step-by-step explanation. Each row of the default matrix has one zero, so 3 non-zeros per row: $s = 3$. A tridiagonal matrix has at most 3 non-zeros per row (diagonal and two neighbours) whatever its size: $s = 3$. 12 and 2998 are the *total* numbers of non-zeros. HHL's original analysis assumes s-sparse matrices so that e^{iAt} can be simulated in time $\mathrm{poly}(s, \log d)$; the notebook does not exploit sparsity (it exponentiates the dense matrix exactly).
Why the other options are incorrect.
(A) A tridiagonal matrix has at most 3 non-zeros per row, not d.
(B) 12 and 2998 are total non-zero counts.
(D) 4 is the row length, but each row has one zero.
Key takeaway. HHL's cost depends on s through Hamiltonian simulation.

Q.10 Single-correct MCQ · Cost of Gaussian elimination

Gaussian elimination on a dense $d \times d$ system costs about $\frac{2}{3}d^3$ floating-point operations. Doubling d from 1000 to 2000 multiplies the work by

(A) ✓ 8 **(B)** 2 **(C)** 4 **(D)** $\frac{2}{3}$

Answer: (A) Detailed solution

Correct answer. **(A)** 8
Concepts required. Cubic cost of Gaussian elimination.
Relevant formulae. work $\approx \frac{2}{3}d^3$.
Step-by-step explanation. Work $\propto d^3$: $(2000/1000)^3 = 2^3 = 8$. For $d = 1000$: $\frac{2}{3} \cdot 10^9 \approx 6.7 \times 10^8$ operations, a fraction of a second today; the cubic growth is what limits direct solvers for very large systems (iterative methods exploit sparsity instead).
Why the other options are incorrect.
(B) 2 would be linear scaling.
(C) 4 would be quadratic scaling.
(D) $\frac{2}{3}$ is the prefactor, not the ratio.
Key takeaway. Direct solvers scale as d^3; sparsity motivates iterative and quantum methods.

Q.11 Single-correct MCQ · Conjugate-gradient scaling

For a sparse symmetric positive-definite matrix with condition number κ, the number of conjugate-gradient iterations needed for a fixed accuracy scales as

(A) $O(\log \kappa)$ **(B)** $O(d^3)$ **(C)** ✓ $O(\sqrt{\kappa})$ **(D)** $O(\kappa^2)$

Answer: (C) Detailed solution

Correct answer. **(C)** $O(\sqrt{\kappa})$
Concepts required. Conjugate-gradient convergence.
Relevant formulae. error $\leq 2(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1})^k \Rightarrow k = O(\sqrt{\kappa}\log\frac{1}{\varepsilon})$.
Step-by-step explanation. Conjugate gradient reduces the error by a factor $\leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$ after k iterations, so $k = O(\sqrt{\kappa}\log(1/\varepsilon))$ iterations suffice; each costs one sparse matrix–vector product, $O(sd)$. Total $O(sd\sqrt{\kappa}\log(1/\varepsilon))$ — the fair classical competitor for HHL, which scales with $\log d$ but polynomially in κ (the original analysis gives κ^2).
Why the other options are incorrect.
(A) $\log\kappa$ is far too optimistic for CG.
(B) d^3 is Gaussian elimination.
(D) κ^2 is HHL's (original) dependence, not CG's.
Key takeaway. The fair classical baseline scales as $sd\sqrt{\kappa}$.

Q.12 SINGLE-CORRECT MCQ · Amplitude encoding

The vector $b = (1, 2, 2, 0)^{\mathsf{T}}$ is amplitude-encoded on two qubits as $|b\rangle = \sum_i b_i |i\rangle / \|b\|$ (index i in binary, first qubit most significant). The probability of measuring $|01\rangle$ is

(A) $\frac{1}{3}$ **(B)** $\frac{2}{9}$ **(C)** $\frac{2}{3}$ **(D)** ✓ $\frac{4}{9}$

Answer: (D) Detailed solution

Correct answer. (D) $\frac{4}{9}$
Concepts required. Amplitude encoding and the Born rule.
Relevant formulae. $|b\rangle = \sum_i b_i |i\rangle / \|b\|$; $P(i) = |b_i|^2 / \|b\|^2$.
Step-by-step explanation. $\|b\| = \sqrt{1+4+4+0} = 3$, so $|b\rangle = \frac{1}{3}(|00\rangle + 2|01\rangle + 2|10\rangle)$. $|01\rangle$ is index 1 with amplitude $\frac{2}{3}$; Born rule: $P = (\frac{2}{3})^2 = \frac{4}{9}$. Amplitude encoding stores d numbers in $\log_2 d$ qubits but loses the norm $\|b\| = 3$, which must be tracked classically (the notebook multiplies by `b_norm` when it recovers x).
Why the other options are incorrect.
(A) $\frac{1}{3}$ is the amplitude of $|00\rangle$.
(B) $\frac{2}{9}$ uses $b_1/\|b\|^2$ without squaring b_1.
(C) $\frac{2}{3}$ is the amplitude, not the probability.
Key takeaway. Amplitude encoding drops the norm; track $\|b\|$ classically.

Q.13 SINGLE-CORRECT MCQ · $|b\rangle$ in the eigenbasis

For the notebook's default $A = H\,\mathrm{diag}(1,2,3,4)H$ (eigenvectors = columns of H) and $b = (1,0,0,0)$, the coefficients $\beta_j = \langle u_j|b\rangle$ of $|b\rangle$ in the eigenbasis are

(A) $(1,0,0,0)$ **(B)** ✓ $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$
(C) $(\frac{1}{2},-\frac{1}{2},\frac{1}{2},-\frac{1}{2})$ **(D)** $(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$

Answer: (B) Detailed solution

Correct answer. (B) $(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$
Concepts required. Expanding b in the eigenbasis.
Relevant formulae. $\beta_j = \langle u_j|b\rangle$; $\sum |\beta_j|^2 = 1$.
Step-by-step explanation. $\beta_j = \langle u_j|b\rangle$ = (column j of H) $\cdot\,(1,0,0,0) = H_{0j} = \frac{1}{2}$ for every j (first row of H). Normalisation: $4 \cdot \frac{1}{4} = 1$ ✓. So $|b\rangle = \frac{1}{2}\sum_j |u_j\rangle$: every eigenvalue contributes with weight $\frac{1}{4}$ — which is why phase estimation produces all four clock values $1,2,3,4$ in the notebook. $(\frac{1}{2},-\frac{1}{2},\dots)$ is the second row of H.
Why the other options are incorrect.
(A) $(1,0,0,0)$ is b in the computational basis, not the eigenbasis.
(C) $(\frac{1}{2},-\frac{1}{2},\dots)$ is the second row of H.
(D) $(\frac{1}{4},\dots)$ are the weights $|\beta_j|^2$, not the amplitudes.
Key takeaway. Equal β_j: all four eigenvalues appear in the clock register.

Paragraph for Questions 14 and 15

The notebook's 2×2 example is $A = \begin{pmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Its eigenvectors are $u_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$ and $u_2 = \frac{1}{\sqrt{2}}(1,-1)^\mathsf{T}$.

Q.14 PARAGRAPH BASED · 2 × 2 example: $x = A^{-1}b$

The solution $x = A^{-1}b$ is

(A) ✓ $(\frac{9}{8},\ \frac{3}{8})^\mathsf{T}$ **(B)** $(1,\ \frac{1}{3})^\mathsf{T}$ **(C)** $(\frac{3}{4},\ \frac{1}{4})^\mathsf{T}$ **(D)** $(\frac{9}{8},\ -\frac{3}{8})^\mathsf{T}$

Answer: (A) Detailed solution

Correct answer. **(A)** $(\frac{9}{8},\ \frac{3}{8})^\mathsf{T}$
Concepts required. 2 × 2 inverse; eigen-decomposition check.
Relevant formulae. $\left(\begin{smallmatrix} a & c \\ c & a \end{smallmatrix}\right)^{-1} = \frac{1}{a^2-c^2}\left(\begin{smallmatrix} a & -c \\ -c & a \end{smallmatrix}\right)$.
Step-by-step explanation. $\det A = 1 - \frac{1}{9} = \frac{8}{9}$, $A^{-1} = \frac{9}{8}\begin{pmatrix} 1 & \frac{1}{3} \\ \frac{1}{3} & 1 \end{pmatrix}$, so $x = \frac{9}{8}(1, \frac{1}{3}) = (\frac{9}{8}, \frac{3}{8})$.

Eigen-route: $b = \frac{1}{\sqrt{2}}(u_1 + u_2)$, $x = \frac{1}{\sqrt{2}}\left(\frac{u_1}{2/3} + \frac{u_2}{4/3}\right) = \frac{1}{2}\left(\frac{3}{2}(1,1) + \frac{3}{4}(1,-1)\right) = (\frac{9}{8}, \frac{3}{8})$. Check: $Ax = (\frac{9}{8} - \frac{1}{8}, -\frac{3}{8} + \frac{3}{8}) = (1, 0)$ ✓. The off-diagonal entry is negative, so A^{-1} has a *positive* off-diagonal entry.
Why the other options are incorrect.
(B) $(1, \frac{1}{3})$ forgets the factor $\frac{1}{\det} = \frac{9}{8}$.
(C) $(\frac{3}{4}, \frac{1}{4})$ is the normalised direction scaled wrongly $(x/\frac{3}{2})$.
(D) $(\frac{9}{8}, -\frac{3}{8})$ has the wrong sign: A^{-1} has a positive off-diagonal entry.
Key takeaway. Verify $Ax = b$ after every inversion.

Q.15 PARAGRAPH BASED · 2 × 2 example: $|x\rangle$

HHL prepares the normalised state $|x\rangle = x/\|x\|$. Its amplitudes, and the probability of measuring $|0\rangle$ on the system qubit, are

(A) $\frac{1}{2}(\sqrt{3}, 1)$ and 0.75 **(B)** $(\frac{9}{8}, \frac{3}{8})$ and $\frac{9}{8}$
(C) ✓ $\frac{1}{\sqrt{10}}(3,1)$ and 0.9 **(D)** $\frac{1}{\sqrt{10}}(3,1)$ and 0.75

Answer: (C) Detailed solution

Correct answer. **(C)** $\frac{1}{\sqrt{10}}(3,1)$ and 0.9
Concepts required. Normalised solution state; Born probabilities.
Relevant formulae. $|x\rangle = x/\|x\|$; $P(0) = |x_0|^2/\|x\|^2$.
Step-by-step explanation. $x = \frac{3}{8}(3,1)$, $\|x\| = \frac{3}{8}\sqrt{10}$, so $|x\rangle = \frac{1}{\sqrt{10}}(3,1) = (0.9487, 0.3162)$. $P(0) = \frac{9}{10} = 0.9$. The overall factor $\frac{3}{8}$ (the norm) is lost in $|x\rangle$; the notebook restores it with $x = \|b\|\, C^{-1} \times$ (amplitudes). $(\frac{9}{8}, \frac{3}{8})$ is not normalised, and a probability above 1 is impossible.
Why the other options are incorrect.
(A) $\frac{1}{2}(\sqrt{3}, 1)$ corresponds to a different direction (ratio $\sqrt{3}$ instead of 3).
(B) $(\frac{9}{8}, \frac{3}{8})$ is not normalised and a probability $\frac{9}{8} > 1$ is impossible.
(D) 0.75 is not $\frac{9}{10}$; the amplitudes $\frac{3}{\sqrt{10}}$ give 0.9.
Key takeaway. HHL outputs the direction of x; the norm needs extra information.

Q.16 SINGLE-CORRECT MCQ · e^{iAt} of a diagonal matrix

For $A = \mathrm{diag}(1, 2)$ and $t = \pi/2$, the matrix e^{iAt} is

(A) $\mathrm{diag}(i,\ 1)$ **(B)** ✓ $\mathrm{diag}(i,\ -1)$ **(C)** $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ **(D)** $\mathrm{diag}(-i,\ -1)$

Answer: (B) Detailed solution

Correct answer. **(B)** $\mathrm{diag}(i,\ -1)$
Concepts required. Exponential of a diagonal (Hermitian) matrix.
Relevant formulae. $e^{iAt} = \sum_j e^{i\lambda_j t} |u_j\rangle \langle u_j|$.
Step-by-step explanation. For a diagonal (or diagonalisable) A, e^{iAt} acts on eigenvectors as $e^{i\lambda t}$: $e^{i\pi/2} = i$ and $e^{i\pi} = -1$. So $\mathrm{diag}(i, -1)$. In general $e^{iAt} = U e^{i\Lambda t} U^\dagger$ — the notebook computes it with `scipy.linalg.expm(1j * A * t)`. The eigenvalues of A become *phases* $e^{i\lambda t}$ of a unitary, which phase estimation can read.
Why the other options are incorrect.
(A) $e^{i\cdot 2\cdot \pi/2} = e^{i\pi} = -1$, not 1.
(C) A diagonal A gives a diagonal e^{iAt}.
(D) $-i$ would be $e^{-i\pi/2}$ (wrong sign of the exponent).
Key takeaway. Eigenvalues become phases $e^{i\lambda t}$ — the input of phase estimation.

Q.17 SINGLE-CORRECT MCQ · Unitarity of e^{iAt}

For which matrices A is e^{iAt} unitary for every real t?

(A) exactly the sparse matrices **(B)** ✓ exactly the Hermitian matrices
(C) exactly the invertible matrices **(D)** exactly the real matrices

Answer: (B) Detailed solution

Correct answer. **(B)** exactly the Hermitian matrices
Concepts required. Unitarity of the evolution operator.
Relevant formulae. $(e^{iAt})^\dagger = e^{-iA^\dagger t} = e^{-iAt}$ iff $A = A^\dagger$.
Step-by-step explanation. $(e^{iAt})^\dagger = e^{-iA^\dagger t}$. This equals $(e^{iAt})^{-1} = e^{-iAt}$ for all t iff $A^\dagger = A$ (differentiate at $t = 0$). Hence HHL's phase-estimation unitary $U = e^{iAt}$ requires a Hermitian A; invertibility is needed separately for A^{-1} to exist. A real non-symmetric matrix gives a non-unitary e^{iAt}.
Why the other options are incorrect.
(A) Sparsity affects cost, not unitarity.
(C) Invertibility does not make e^{iAt} unitary.
(D) A real non-symmetric A gives a non-unitary e^{iAt}.
Key takeaway. Hermitian $A \Leftrightarrow$ unitary e^{iAt} for all t.

Q.18 Single-correct MCQ · Bounds on $\|A^{-1}b\|$

For the default system (A with eigenvalues $1, 2, 3, 4$ and $\|b\| = 1$), the norm $\|x\| = \|A^{-1}b\|$ is 0.5966. Which bound must every such norm satisfy?

(A) ✓ $\dfrac{\|b\|}{\lambda_{\max}} \le \|x\| \le \dfrac{\|b\|}{\lambda_{\min}}$, i.e. $0.25 \le 0.5966 \le 1$

(B) $\|x\| = \|b\|/\kappa = 0.25$ exactly

(C) $\|x\| \le \|b\|/\lambda_{\max} = 0.25$

(D) $\|b\|\,\lambda_{\min} \le \|x\| \le \|b\|\,\lambda_{\max}$, i.e. $1 \le \|x\| \le 4$

Answer: (A) Detailed solution

Correct answer. **(A)** $\dfrac{\|b\|}{\lambda_{\max}} \le \|x\| \le \dfrac{\|b\|}{\lambda_{\min}}$, i.e. $0.25 \le 0.5966 \le 1$

Concepts required. Norm bounds for $A^{-1}b$.

Relevant formulae. $\|b\|/\lambda_{\max} \le \|A^{-1}b\| \le \|b\|/\lambda_{\min}$ (Hermitian, positive).

Step-by-step explanation. $\|x\|^2 = \sum_j |\beta_j|^2/\lambda_j^2$ lies between $\|b\|^2/\lambda_{\max}^2$ and $\|b\|^2/\lambda_{\min}^2$. Here $\|x\|^2 = \frac{1}{4}(1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16}) = 0.3559$, $\|x\| = 0.5966$, between 0.25 and 1. In HHL $\|x\|^2$ times C^2 is exactly the success probability (with $C = \lambda_{\min} = 1$: 0.3559, the value recorded by the notebook).

Why the other options are incorrect.

(B) $\|b\|/\kappa$ is not a formula for $\|x\|$; here $\|x\| = 0.5966 \neq 0.25$.

(C) 0.25 is the *lower* bound, not an upper bound.

(D) Multiplies by the eigenvalues; that bounds $\|Ab\|$, not $\|A^{-1}b\|$.

Key takeaway. $C^2\|x\|^2/\|b\|^2$ is HHL's success probability.

Q.19 Single-correct MCQ · Singular matrices

For which values of a is $\begin{pmatrix} a & 1 \\ 1 & a \end{pmatrix}$ singular (so that HHL cannot be applied)?

(A) $a = \pm 2$ **(B)** ✓ $a = \pm 1$ **(C)** $a = 0$ **(D)** no real a

Answer: (B) Detailed solution

Correct answer. **(B)** $a = \pm 1$

Concepts required. Singular matrices have a zero eigenvalue.

Relevant formulae. $\det\left(\begin{smallmatrix} a & 1 \\ 1 & a \end{smallmatrix}\right) = a^2 - 1$.

Step-by-step explanation. Eigenvalues $a \pm 1$; the matrix is singular iff one of them is 0: $a = 1$ (eigenvalue $a - 1 = 0$) or $a = -1$ (eigenvalue $a + 1 = 0$). Equivalently $\det = a^2 - 1 = 0$. For $a = 0$ the eigenvalues are ± 1 (invertible, $\kappa = 1$, but indefinite — HHL handles negative eigenvalues via two's complement clock values). The notebook rejects singular matrices (`det` $< 10^{-12}$ or $\kappa > 10^{12}$).

Why the other options are incorrect.

(A) $a = \pm 2$ gives eigenvalues $\{1, 3\}$ or $\{-3, -1\}$: invertible.

(C) $a = 0$ gives eigenvalues ± 1: invertible.

(D) $a = \pm 1$ are real values that make it singular.

Key takeaway. HHL requires invertibility; the notebook checks it.

Q.20 Assertion–Reason · Scaling and κ

Assertion (A): Multiplying an invertible matrix by a non-zero constant c does not change its condition number.

Reason (R): Multiplying A by c multiplies every eigenvalue by c^2.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) ✓ A is true, but R is false

(D) A is false, but R is true

Answer: (C) Detailed solution

Correct answer. **(C)** A is true, but R is false
Concepts required. Scaling and the condition number.
Relevant formulae. $\lambda_i(cA) = c\lambda_i(A)$; $\kappa(cA) = \kappa(A)$.
Step-by-step explanation. **A is true:** all singular values are multiplied by $|c|$, so their ratio is unchanged. **R is false:** eigenvalues of cA are $c\lambda_i$ (multiply $Av = \lambda v$ by c), not $c^2\lambda_i$. Scaling still matters for HHL: the choice of t and C is relative to the eigenvalues (the notebook's `"auto"` time rescales automatically). Answer: (C).
Why the other options are incorrect.
(A) needs A true, R true *and* R explaining A; R is false.
(B) needs A and R both true but R not explaining A; R is false.
(D) needs A false and R true; A is true and R is false.
Key takeaway. Scale changes t and C, not κ.

SILICO FELLER

SECTION B · Quantum Computing Foundations Q.21–40

Q.21 SINGLE-CORRECT MCQ · Exact phase estimation

Phase estimation with 4 counting qubits is applied to an eigenstate of U with eigenvalue $e^{2\pi i \cdot 3/16}$. The measured counting register is

(A) $|0011\rangle$ with probability $\frac{3}{16}$

(B) $|0011\rangle$ or $|0100\rangle$, each with probability $\frac{1}{2}$

(C) ✓ $|0011\rangle$ with probability 1

(D) $|1100\rangle$ with probability 1

Answer: (C) Detailed solution

Correct answer. (C) $|0011\rangle$ with probability 1

Concepts required. Phase estimation with an exactly representable phase.

Relevant formulae. $\varphi = k/2^t$: counting register $=$ QFT $|k\rangle \to$ inverse QFT gives $|k\rangle$ with probability 1.

Step-by-step explanation. When $\varphi = k/2^t$ is exactly representable, the controlled powers produce $\frac{1}{\sqrt{16}} \sum_j e^{2\pi i j k/16} |j\rangle = \text{QFT}\,|k\rangle$, and the inverse QFT returns exactly $|k\rangle$. Here $k = 3 = 0011_2$ (most significant bit first): probability 1. $|1100\rangle$ is the bit-reversed string (the swaps of the QFT fix the order). In the notebook's default every eigenvalue maps to such an exact clock value, which is why its fidelity is 1.

Why the other options are incorrect.

(A) Probability $\frac{3}{16}$ confuses the phase with a probability.

(B) Splitting over two outcomes happens only for non-representable phases.

(D) $|1100\rangle$ is 3 written in reverse bit order.

Key takeaway. Exact phases give certain outcomes — the basis of the notebook's fidelity 1.

Q.22 SINGLE-CORRECT MCQ · Clock value of an eigenvalue

In HHL, phase estimation uses $U = e^{iAt}$, so an eigenvalue λ gives the clock value $k = \lambda t\, 2^{n_c}/(2\pi)$. With $n_c = 4$ and $t = \pi/8$ (the notebook's default), the eigenvalue $\lambda = 4$ gives

(A) $k = 2$

(B) $k = 4\pi$

(C) $k = 8$

(D) ✓ $k = 4$, i.e. $|0100\rangle$

Answer: (D) Detailed solution

Correct answer. (D) $k = 4$, i.e. $|0100\rangle$

Concepts required. Clock value of an eigenvalue in HHL.

Relevant formulae. $k = \lambda t\, 2^{n_c}/(2\pi)$.

Step-by-step explanation. $e^{i\lambda t} = e^{2\pi i \varphi}$ with $\varphi = \lambda t/(2\pi)$; the clock reads $k = \varphi\, 2^{n_c}$. So $k = 4 \cdot \frac{\pi}{8} \cdot \frac{16}{2\pi} = 4$. With $t = \pi/8$ every eigenvalue λ is read as $k = \lambda$ (the `"auto"` time maps $|\lambda|_{\min} = 1$ to clock value 1), so the default eigenvalues $1, 2, 3, 4$ give the exact clock values $1, 2, 3, 4$ printed by the notebook.

Why the other options are incorrect.

(A) 2 halves the factor $t\, 2^{n_c}/(2\pi) = 1$: it would need $t = \pi/16$.

(B) 4π forgets to divide by 2π.

(C) 8 doubles the factor $t\, 2^{n_c}/2\pi = 1$.

Key takeaway. With $t = \pi/8$, $n_c = 4$: clock value = eigenvalue.

Q.23 MATCHING LISTS · Stages of phase estimation

Match each step of quantum phase estimation (List-I) with its effect (List-II).

List-I		List-II	
(P)	Hadamards on the counting register	**(1)**	converts the pattern of phases $e^{2\pi i 2^j \varphi}$ into the binary number $\approx 2^t\varphi$
(Q)	controlled U^{2^j} gates	**(2)**	returns one estimate y of $2^t\varphi$
(R)	inverse QFT on the counting register	**(3)**	creates the uniform superposition $\frac{1}{\sqrt{2^t}}\sum_j \lvert j\rangle$
(S)	measurement of the counting register	**(4)**	write the phase $e^{2\pi i 2^j \varphi}$ onto counting qubit j (phase kickback)

(A) P→3, Q→1, R→4, S→2
(B) P→3, Q→4, R→2, S→1
(C) P→4, Q→3, R→1, S→2
(D) ✓ P→3, Q→4, R→1, S→2

Answer: (D) Detailed solution

Correct answer. **(D)** P→3, Q→4, R→1, S→2
Concepts required. The four stages of phase estimation.
Relevant formulae. $\frac{1}{\sqrt{2^t}}\sum_j e^{2\pi i j\varphi}\lvert j\rangle \otimes \lvert u\rangle \xrightarrow{\text{QFT}^{-1}} \approx \lvert 2^t\varphi\rangle\lvert u\rangle$.
Step-by-step explanation.

- P: Hadamards give the uniform superposition (3).
- Q: controlled U^{2^j} on the eigenstate kicks the phase back to the control (4).
- R: the counting register is then QFT $\lvert k\rangle$ (exactly or approximately); the inverse QFT turns it into $\lvert k\rangle$ (1).
- S: measurement gives one estimate (2). In HHL this measurement is *not* done: the clock register stays coherent for the eigenvalue inversion.

Why the other options are incorrect.
(A) Q→1, R→4 swaps the kickback and the inverse QFT.
(B) R→2, S→1 swaps the inverse QFT and the measurement.
(C) P→4, Q→3 swaps Hadamards and controlled powers.
Key takeaway. HHL runs phase estimation without the final measurement.

Q.24 SINGLE-CORRECT MCQ · Phase kickback

A control qubit in $\lvert+\rangle$ and a target in an eigenstate $\lvert u\rangle$ of U ($U\lvert u\rangle = e^{i\phi}\lvert u\rangle$) pass through a controlled-U. The result is

(A) ✓ $\frac{1}{\sqrt{2}}(\lvert 0\rangle + e^{i\phi}\lvert 1\rangle)\otimes\lvert u\rangle$
(B) $\frac{1}{\sqrt{2}}(\lvert 0\rangle\lvert u\rangle + \lvert 1\rangle\lvert u'\rangle)$ with $\lvert u'\rangle$ orthogonal to $\lvert u\rangle$
(C) $\frac{1}{\sqrt{2}}(e^{i\phi}\lvert 0\rangle + \lvert 1\rangle)\otimes\lvert u\rangle$
(D) $\lvert+\rangle\otimes e^{i\phi}\lvert u\rangle$

Answer: (A) Detailed solution

Correct answer. **(A)** $\frac{1}{\sqrt{2}}(\lvert 0\rangle + e^{i\phi}\lvert 1\rangle)\otimes\lvert u\rangle$
Concepts required. Phase kickback.

Relevant formulae. $cU\,|+\rangle\,|u\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}\,|1\rangle)\,|u\rangle$.
Step-by-step explanation. $\frac{1}{\sqrt{2}}(|0\rangle\,|u\rangle + |1\rangle\,U\,|u\rangle) = \frac{1}{\sqrt{2}}(|0\rangle\,|u\rangle + e^{i\phi}\,|1\rangle\,|u\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}\,|1\rangle)\,|u\rangle$. The target is unchanged; the phase appears as a *relative* phase on the control (phase kickback). $|+\rangle \otimes e^{i\phi}\,|u\rangle$ would be a global phase — unobservable. This is how phase estimation extracts eigenvalues without disturbing the eigenvector.
Why the other options are incorrect.
(B) An eigenstate is not changed into an orthogonal state.
(C) The phase multiplies the $|1\rangle$ component, not $|0\rangle$.
(D) A global phase on the target would be unobservable; the phase moves to the control.
Key takeaway. Controlled unitaries write eigenphases onto the control qubit.

Q.25 SINGLE-CORRECT MCQ · Inverse QFT of QFT $|1\rangle$

The inverse QFT on two qubits is applied to $\frac{1}{2}(\,|00\rangle + i\,|01\rangle - |10\rangle - i\,|11\rangle\,)$. The result is

(A) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ **(B)** ✓ $|01\rangle$
(C) $|11\rangle$ **(D)** $|10\rangle$

Answer: (B) Detailed solution

Correct answer. **(B)** $|01\rangle$
Concepts required. QFT of a basis state; inverse QFT.
Relevant formulae. $\mathrm{QFT}\,|k\rangle = \frac{1}{\sqrt{N}}\sum_y \omega^{ky}\,|y\rangle$, $\omega = i$ for $N = 4$.
Step-by-step explanation. $\mathrm{QFT}\,|k\rangle = \frac{1}{2}\sum_y i^{ky}\,|y\rangle$ ($\omega = e^{2\pi i/4} = i$). For $k = 1$: $\frac{1}{2}(1, i, i^2, i^3) = \frac{1}{2}(1, i, -1, -i)$ — exactly the given state. So QFT^{-1} returns $|1\rangle = |01\rangle$. ($k = 3$ would give $\frac{1}{2}(1, -i, -1, i)$.) This is the last step of phase estimation, the link between HHL and the QFT lecture.
Why the other options are incorrect.
(A) The uniform state is $\mathrm{QFT}\,|0\rangle$.
(C) $|11\rangle$ ($k = 3$) would need $\frac{1}{2}(1, -i, -1, i)$.
(D) $|10\rangle$ ($k = 2$) would need $\frac{1}{2}(1, -1, 1, -1)$.
Key takeaway. Recognise $\mathrm{QFT}\,|k\rangle$ by the phase ratio ω^k between neighbours.

Q.26 SINGLE-CORRECT MCQ · QFT gate counts in HHL

The notebook builds the inverse QFT (in phase estimation) and the QFT (in the uncomputation) on $n_c = 4$ clock qubits from controlled phase gates `cr1` and `swaps`. How many `cr1` and `swap` gates does the whole HHL circuit contain?

(A) 12 and 8 **(B)** 16 and 4 **(C)** 6 and 2 **(D)** ✓ 12 and 4

Answer: (D) Detailed solution

Correct answer. **(D)** 12 and 4
Concepts required. Gate counts of the QFT.
Relevant formulae. QFT_n: n H, $\frac{n(n-1)}{2}$ controlled rotations, $\lfloor n/2 \rfloor$ swaps.
Step-by-step explanation. One QFT on n qubits has $\binom{n}{2} = n(n-1)/2$ controlled rotations and $\lfloor n/2 \rfloor$ swaps: for $n = 4$, 6 and 2. HHL needs the inverse QFT once and the QFT once (uncomputation): 12 `cr1` and 4 `swap` — exactly the recorded statistics (`cr1: 12, swap: 4`).
Why the other options are incorrect.
(A) Each QFT has only 2 swaps for 4 qubits.

(B) 16 counts the Hadamards (4×4), not the `cr1`.

(C) 6 and 2 count only one QFT; HHL needs two.

Key takeaway. The QFT parts of HHL are cheap compared with the controlled powers.

Q.27 SINGLE-CORRECT MCQ · $R_Y(2\arcsin c)$

An ancilla in $|0\rangle$ is rotated by $R_Y(\theta) = e^{-i\theta Y/2}$ with $\theta = 2\arcsin(\frac{1}{2})$. The probability of then measuring $|1\rangle$ is

(A) $\frac{3}{4}$ **(B)** ✓ $\frac{1}{4}$ **(C)** $\sin^2(\frac{1}{2}) \approx 0.23$ **(D)** $\frac{1}{2}$

Answer: (B) Detailed solution

Correct answer. **(B)** $\frac{1}{4}$

Concepts required. R_Y rotation amplitude.

Relevant formulae. $R_Y(2\arcsin c)\,|0\rangle = \sqrt{1-c^2}\,|0\rangle + c\,|1\rangle$.

Step-by-step explanation. $R_Y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$. With $\theta = 2\arcsin c$: $\sin\frac{\theta}{2} = c = \frac{1}{2}$, so the amplitude of $|1\rangle$ is $\frac{1}{2}$ and $P(1) = \frac{1}{4}$; $\theta = \pi/3$. This is HHL's eigenvalue inversion: $c = C/\lambda$ (here $C = 1$, $\lambda = 2$) puts amplitude C/λ on the ancilla's $|1\rangle$.

Why the other options are incorrect.

(A) $\frac{3}{4} = \cos^2\frac{\pi}{6}$ is $P(0)$, the probability of the other outcome.

(C) $\sin^2(\frac{1}{2})$ treats $\frac{1}{2}$ as an angle.

(D) $\frac{1}{2}$ is the amplitude, not the probability.

Key takeaway. The rotation angle encodes an amplitude $c = C/\lambda$.

Q.28 SINGLE-CORRECT MCQ · Polarity X gates

To apply a rotation only when the 4-qubit clock register is in $|k\rangle$, the notebook surrounds a fully controlled `cry` with X gates on the clock qubits whose bit of k is 0 (clock[0] = most significant bit). For $k = 5$, the X gates act on

(A) clock[2] and clock[3] **(B)** ✓ clock[0] and clock[2]

(C) clock[0] only **(D)** clock[1] and clock[3]

Answer: (B) Detailed solution

Correct answer. **(B)** clock[0] and clock[2]

Concepts required. Selecting a basis state with polarity X gates.

Relevant formulae. zero bits of k (MSB first) get X before and after the all-ones-controlled gate.

Step-by-step explanation. $5 = 0101_2$: bits (clock[0], clock[1], clock[2], clock[3]) $= (0, 1, 0, 1)$. The zero bits are clock[0] and clock[2]; X on them maps $|0101\rangle$ to $|1111\rangle$, which triggers the all-ones control, and the second pair of X gates restores the register. Summed over $k = 1, \ldots, 15$ these "polarity" gates account for the 56 `x` gates in the recorded statistics (twice the 28 zero bits).

Why the other options are incorrect.

(A) clock[2] and clock[3]: clock[3] is a one bit.

(C) clock[0] alone would select 1101 or similar.

(D) clock[1] and clock[3] are the *one* bits of 0101.

Key takeaway. X on zero bits turns "control on $|k\rangle$" into "control on all ones".

Q.29 Statement I / II · Controlled powers U^{2^j}

Statement I: A controlled-U^{2^j} can be implemented as 2^j consecutive controlled-U gates.
Statement II: For $U = e^{iAt}$, $U^{2^j} = e^{iAt\,2^j}$.

(A) ✓ Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Statement I and Statement II are true
Concepts required. Powers of controlled unitaries.
Relevant formulae. $(\mathrm{c}U)^m = \mathrm{c}(U^m)$; $(e^{iAt})^m = e^{iAmt}$.
Step-by-step explanation. **I is true:** the control condition is the same for every factor, so $(cU)^{2^j} = c(U^{2^j})$ — but the cost doubles with j, which is why efficient Hamiltonian simulation is essential. **II is true:** A commutes with itself, so $(e^{iAt})^m = e^{iAmt}$. The notebook registers each U^{2^j} directly as a matrix (`np.linalg.matrix_power(U, 2**j)`). Answer: (A).
Why the other options are incorrect.
(B) needs both statements false; both are true.
(C) needs I true and II false; II is true.
(D) needs I false and II true; I is true.
Key takeaway. Phase estimation needs U^{2^j}; its cost drives HHL's cost.

Q.30 Matching lists · Two's complement clock values

HHL reads the 4-qubit clock as a signed (two's complement) number: values $k \geq 8$ mean $k - 16$. Match each clock value (List-I) with the signed value (List-II).

List-I		List-II	
(P)	$k = 3$	**(1)**	-1
(Q)	$k = 8$	**(2)**	-4
(R)	$k = 12$	**(3)**	3
(S)	$k = 15$	**(4)**	-8

(A) P→3, Q→2, R→4, S→1
(B) P→3, Q→4, R→1, S→2
(C) ✓ P→3, Q→4, R→2, S→1
(D) P→1, Q→4, R→2, S→3

Answer: (C) Detailed solution

Correct answer. **(C)** P→3, Q→4, R→2, S→1
Concepts required. Two's complement on the clock register.
Relevant formulae. signed $= k - 2^{n_c}$ for $k \geq 2^{n_c-1}$.
Step-by-step explanation. Rule: signed $= k$ for $k < 8$, $k - 16$ for $k \geq 8$.

- P: $3 < 8$: 3 (3). Q: $8 - 16 = -8$ (4), the most negative value. R: $12 - 16 = -4$ (2). S: $15 - 16 = -1$ (1).

A negative eigenvalue λ gives $e^{i\lambda t}$ with phase $1 + \lambda t/2\pi$ (mod 1), near the top of the clock range; the notebook uses `signed = k_val - 2**nc if k_val >= 2**(nc - 1) else k_val`.
Why the other options are incorrect.
(A) Q→2, R→4 gives $k = 8$ the value -4 and $k = 12$ the value -8; subtract 16 each time.

(B) R→1, S→2 swaps $k = 12$ (-4) and $k = 15$ (-1).

(D) P→1 makes $k = 3$ negative; values below 8 are positive.

Key takeaway. Negative eigenvalues live at the top of the clock range.

Q.31 Single-correct MCQ · Post-selection

A state $|\Psi\rangle = \sqrt{1-p}\,|\psi_0\rangle\,|0\rangle + \sqrt{p}\,|\psi_1\rangle\,|1\rangle$ (normalised $|\psi_0\rangle$, $|\psi_1\rangle$) is measured on the last qubit and the result 1 is kept (post-selection). Which statement is correct?

(A) ✓ The outcome 1 occurs with probability p and leaves the rest in $|\psi_1\rangle$

(B) The outcome 1 occurs with probability $\sqrt{p}$

(C) Post-selection succeeds with probability 1 if $p > 0$

(D) The remaining state is $\sqrt{p}\,|\psi_1\rangle$, not normalised

Answer: (A) Detailed solution

Correct answer. **(A)** The outcome 1 occurs with probability p and leaves the rest in $|\psi_1\rangle$
Concepts required. Post-selection.
Relevant formulae. $P(1) = \|\sqrt{p}\,|\psi_1\rangle\,\|^2 = p$; post-measurement state renormalised.
Step-by-step explanation. Born rule: $P(1) = \|\sqrt{p}\,|\psi_1\rangle\,\|^2 = p$; the post-measurement state is renormalised: $|\psi_1\rangle$. On average $1/p$ repetitions are needed. In HHL $p = P_{\text{success}} = C^2\|A^{-1}b\|^2$ and $|\psi_1\rangle = |x\rangle$; the notebook recorded $p = 0.3559$ (7086 accepted of 20000 shots).
Why the other options are incorrect.

(B) Probabilities are squared amplitudes: p, not $\sqrt{p}$.

(C) Success has probability p, not 1.

(D) The post-measurement state is renormalised to $|\psi_1\rangle$.

Key takeaway. Expected repetitions $1/p$; HHL's $p = C^2\|A^{-1}b\|^2$.

Q.32 Single-correct MCQ · System–clock entanglement

After phase estimation in HHL the system and clock registers are in $\sum_j \beta_j\,|u_j\rangle\,|k_j\rangle$ (distinct k_j). When is this a product state of the two registers?

(A) always

(B) never

(C) ✓ when $|b\rangle$ is an eigenvector of A (only one $\beta_j \neq 0$)

(D) when all β_j are equal

Answer: (C) Detailed solution

Correct answer. **(C)** when $|b\rangle$ is an eigenvector of A (only one $\beta_j \neq 0$)
Concepts required. Entanglement between system and clock.
Relevant formulae. Schmidt decomposition $\sum_j \beta_j\,|u_j\rangle\,|k_j\rangle$ with orthonormal $|k_j\rangle$.
Step-by-step explanation. With distinct clock values, the Schmidt coefficients are the $|\beta_j|$; the state is a product iff only one is non-zero, i.e. $|b\rangle$ is an eigenvector. Otherwise system and clock are entangled — which is why the clock must be uncomputed after the inversion: leaving it entangled would leave the system in a mixture rather than $|x\rangle$. Equal β_j (the notebook's default) give *maximal* entanglement over the four eigenvalues.
Why the other options are incorrect.

(A) Not always: several non-zero β_j give entanglement.

(B) Not never: an eigenvector $|b\rangle$ gives a product state.

(D) Equal β_j give maximal, not zero, entanglement.

Key takeaway. This entanglement is why the clock must be uncomputed.

Q.33 SINGLE-CORRECT MCQ · Memory of the HHL state

A state vector of n qubits needs 16×2^n bytes (fp64). The notebook's default HHL circuit has 7 qubits, yet it prints "Estimated memory: 4.10 KB". Why?

(A) The density matrix of 7 qubits needs exactly 4.10 KB

(B) Each qubit needs $16 \times 7 = 112$ B and there are 36 of them

(C) ✓ The state needs 2048 B and the notebook reserves one extra copy (for reading the full state), giving 4096 B

(D) Custom gates add 2048 B of matrix storage per qubit

Answer: (C) Detailed solution

Correct answer. **(C)** The state needs 2048 B and the notebook reserves one extra copy (for reading the full state), giving 4096 B

Concepts required. State-vector memory; the notebook's memory check.

Relevant formulae. 16×2^n B; `check_memory(..., extra_copies=1)` adds one copy on CPU.

Step-by-step explanation. $16 \times 2^7 = 2048$ B. The memory check is called with `extra_copies=1` because `cudaq.get_state` returns a second copy of the state: $2 \times 2048 = 4096$ B $= 4.10$ KB. A density matrix would need $16 \times 4^7 = 262\,144$ B. Even with 8 clock qubits (11 qubits) the state needs only 32 kB: HHL simulations of small systems are cheap; the circuit depth, not memory, grows quickly (Section D).

Why the other options are incorrect.

(A) A 7-qubit density matrix needs 262 144 B.

(B) Memory grows as 2^n, not $16n$ per qubit.

(D) Custom-gate matrices are not counted by the memory check.

Key takeaway. Small HHL simulations are cheap in memory.

Q.34 SINGLE-CORRECT MCQ · Cost of state preparation

Preparing an arbitrary normalised n-qubit amplitude vector $|b\rangle$ from $|0\ldots0\rangle$ with one- and two-qubit gates requires in general

(A) $O(n)$ gates

(B) $O(\log n)$ gates

(C) a single gate per qubit

(D) ✓ $O(2^n)$ gates

Answer: (D) Detailed solution

Correct answer. **(D)** $O(2^n)$ gates

Concepts required. Cost of general state preparation.

Relevant formulae. $2^{n+1} - 2$ real parameters $\Rightarrow \Omega(2^n)$ gates in general.

Step-by-step explanation. An arbitrary state has $2^{n+1} - 2$ real degrees of freedom, and each gate fixes only $O(1)$ of them, so $\Omega(2^n)$ gates are needed in general (standard constructions use $O(2^n)$ uniformly controlled rotations). HHL's exponential advantage therefore requires $|b\rangle$ to be *efficiently* preparable (structured b, or qRAM). The notebook side-steps this on the simulator with `kernel.qalloc(vector)`, which initialises the amplitudes directly — not possible on hardware.

Why the other options are incorrect.

(A) $O(n)$ gates suffice only for special states.

(B) $O(\log n)$ is far too few for a generic state.

(C) One gate per qubit makes only product states.

Key takeaway. Efficient $|b\rangle$ preparation is an assumption of HHL's speed-up.

Q.35 Assertion–Reason · Measuring the clock too early

Assertion (A): In HHL, measuring the clock register immediately after phase estimation would destroy the superposition over eigencomponents that $|x\rangle$ needs.

Reason (R): The inverse QFT is a unitary operation.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) ✓ Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (B) Detailed solution

Correct answer. **(B)** Both A and R are true, but R is *not* the correct explanation of A
Concepts required. Measurement collapse vs. unitarity.
Relevant formulae. measuring $|k_j\rangle$ leaves $|u_j\rangle\,|k_j\rangle$ with probability $|\beta_j|^2$.
Step-by-step explanation. **A is true:** measuring the clock in $\sum_j \beta_j\,|u_j\rangle\,|k_j\rangle$ collapses it to one term $|u_j\rangle\,|k_j\rangle$ with probability $|\beta_j|^2$; the relative weights β_j/λ_j of $|x\rangle$ are lost. **R is true** but does not explain A: the reason is the collapse caused by measurement, not the unitarity of the inverse QFT. Answer: (B).
Why the other options are incorrect.

(A) needs A true, R true *and* R explaining A; R does not explain A.

(C) needs A true and R false; R is true.

(D) needs A false and R true; A is true.

Key takeaway. Keep the clock coherent until uncomputation.

Q.36 Single-correct MCQ · Swap test

A swap test between states $|\psi\rangle$ and $|\phi\rangle$ gives the ancilla outcome 0 with probability $P(0) = \frac{1}{2}(1 + |\langle\psi|\phi\rangle|^2)$. If $P(0) = 0.9$ is measured, the overlap $|\langle\psi|\phi\rangle|^2$ is

(A) ✓ 0.8 **(B)** 0.95 **(C)** 0.9 **(D)** 0.4

Answer: (A) Detailed solution

Correct answer. **(A)** 0.8
Concepts required. Swap test.
Relevant formulae. $P(0) = \frac{1}{2}(1 + |\langle\psi|\phi\rangle|^2)$.
Step-by-step explanation. $|\langle\psi|\phi\rangle|^2 = 2P(0) - 1 = 0.8$. Swap tests are one way to extract information from HHL's output $|x\rangle$ (e.g. its overlap with a reference) without reading all amplitudes. The notebook instead reads the simulated state directly and reports the fidelity $|\langle x_{\text{classical}}|x_{\text{quantum}}\rangle|^2$.
Why the other options are incorrect.

(B) 0.95 has no derivation from the formula.

(C) 0.9 is $P(0)$ itself.

(D) 0.4 uses $P(0) - \frac{1}{2}$ without doubling.

Key takeaway. Overlaps of $|x\rangle$ with references are measurable without tomography.

Paragraph for Questions 37 and 38

Quantum phase estimation with $t = 2$ counting qubits is applied to an eigenstate $|u\rangle$ of a unitary U, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$. The counting register ends in $\sum_y \alpha_y |y\rangle$ with $\alpha_y = \frac{1}{4}\sum_{j=0}^{3} e^{2\pi i j(\varphi - y/4)}$, and the outcome y estimates $\varphi \approx y/4$.

Q.37 PARAGRAPH BASED · QPE with $\varphi = 1/4$

For $\varphi = \frac{1}{4}$, the measured counting register is

(A) $|10\rangle$ with probability 1
(B) ✓ $|01\rangle$ with probability 1
(C) $|01\rangle$ with probability $\frac{1}{4}$
(D) $|00\rangle$ with probability 1

Answer: (B) Detailed solution

Correct answer. (B) $|01\rangle$ with probability 1
Concepts required. Exact phase estimation.
Relevant formulae. $\alpha_y = \frac{1}{M}\sum_j e^{2\pi i j(\varphi - y/M)} = \delta_{y,M\varphi}$ for $M\varphi \in \mathbb{Z}$.
Step-by-step explanation. $\varphi = \frac{1}{4} = 1/2^2$ is exact: $\alpha_y = \frac{1}{4}\sum_j e^{2\pi i j(1/4 - y/4)}$ is 1 for $y = 1$ and 0 otherwise (geometric sum of roots of unity). So $y = 1 = |01\rangle$ with certainty, $\varphi = y/4 = \frac{1}{4}$ exactly.
Why the other options are incorrect.
(A) $|10\rangle$ represents $\varphi = \frac{1}{2}$.
(C) An exact phase gives probability 1, not $\frac{1}{4}$.
(D) $|00\rangle$ represents $\varphi = 0$.
Key takeaway. Exact phases are measured with certainty.

Q.38 PARAGRAPH BASED · QPE leakage, $\varphi = 1/3$

For $\varphi = \frac{1}{3}$, the most likely outcome and its probability are

(A) all four outcomes, 0.25 each
(B) $|01\rangle$, exactly 1
(C) $|10\rangle$, about 0.70
(D) ✓ $|01\rangle$, about 0.70

Answer: (D) Detailed solution

Correct answer. (D) $|01\rangle$, about 0.70
Concepts required. Phase-estimation leakage for non-representable phases.
Relevant formulae. $P(y) = \left|\frac{1}{M}\sum_j e^{2\pi i j(\varphi - y/M)}\right|^2$; nearest outcome $\geq 4/\pi^2$.
Step-by-step explanation. $\frac{1}{3} = 0.0101\ldots_2$ is not a 2-bit fraction. $P(y) = |\alpha_y|^2$ evaluates to $(0.0625, 0.6998, 0.1875, 0.0502)$ for $y = 0, 1, 2, 3$: the nearest value $y = 1$ ($\varphi \approx 0.25$) wins with ≈ 0.70, but the rest leaks to neighbours. (The general guarantee for the nearest outcome is $\geq 4/\pi^2 \approx 0.405$.) In HHL such leakage spreads an eigenvalue over several clock values, each inverted differently — the fidelity drops below 1 (e.g. 0.953 for the default with only 3 clock qubits).
Why the other options are incorrect.
(A) The distribution is peaked, not uniform.
(B) Probability 1 needs an exactly representable phase.
(C) $|10\rangle$ (0.5) is farther from $\frac{1}{3}$ than $|01\rangle$ (0.25).
Key takeaway. Leakage is HHL's main source of error in small simulations.

Q.39 Statement I / II · QFT unitarity and cost

Statement I: The inverse QFT is the adjoint of the QFT, so applying it after the QFT returns the original state.

Statement II: The QFT on n qubits requires a number of gates exponential in n.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. **(C)** Statement I is true, but Statement II is false
Concepts required. QFT unitarity and gate count.
Relevant formulae. $\mathrm{QFT}^{-1} = \mathrm{QFT}^{\dagger}$; $O(n^2)$ gates.
Step-by-step explanation. **I is true:** the QFT is unitary, $\mathrm{QFT}^{-1} = \mathrm{QFT}^{\dagger}$ (reverse the gate order and conjugate the angles — the notebook's two QFT blocks are mirror images). **II is false:** n Hadamards, $n(n-1)/2$ controlled rotations and $\lfloor n/2 \rfloor$ swaps: $O(n^2)$ gates. Phase estimation's cost is dominated by the controlled powers U^{2^j}, not by the QFT. Answer: (C).
Why the other options are incorrect.

(A) needs both statements true; II is false.

(B) needs both statements false; I is true.

(D) needs I false and II true; I is true and II is false.

Key takeaway. The QFT is cheap; the controlled powers are not.

Q.40 Single-correct MCQ · Shift $A \to A + I$

With $n_c = 4$ and $t = \pi/8$, the matrix A is replaced by $A + I$. How do the ideal clock values change?

(A) ✓ each clock value increases by 1

(B) each clock value increases by $\pi/8$

(C) they are unchanged, because I only adds a global phase

(D) each clock value doubles

Answer: (A) Detailed solution

Correct answer. **(A)** each clock value increases by 1
Concepts required. Controlled global phases become relative phases.
Relevant formulae. $e^{i(A+cI)t} = e^{ict}e^{iAt}$; shift of clock values $ct\,2^{n_c}/(2\pi)$.
Step-by-step explanation. $e^{i(A+I)t} = e^{it}e^{iAt}$ (since I commutes with A). Without control this would be a global phase, but in *controlled-U* it becomes a relative phase on the control: every eigenvalue λ becomes $\lambda + 1$ and every clock value $k = \lambda t\,2^{n_c}/(2\pi)$ increases by $t\,2^{n_c}/(2\pi) = 1$. Global phases of a gate matter once the gate is controlled — phase estimation measures exactly these.
Why the other options are incorrect.

(B) $\pi/8$ is t, not the shift in clock units.

(C) A global phase of U becomes a relative phase once U is controlled.

(D) Adding I shifts, it does not scale.

Key takeaway. Phase estimation measures eigenvalues including constant shifts.

SECTION C · HHL Algorithm Fundamentals Q.41–60

Q.41 Single-correct MCQ · State after inversion

After phase estimation and the controlled rotation (before uncomputation), the ideal HHL state is (system ⊗ clock ⊗ ancilla)

(A) $\sum_j \beta_j |u_j\rangle |0\rangle \left(\sqrt{1-\frac{C^2}{\lambda_j^2}}\,|0\rangle + \frac{C}{\lambda_j}|1\rangle\right)$ **(B)** $\sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle |0\rangle |1\rangle$

(C) $\sum_j \beta_j |u_j\rangle |k_j\rangle \left(\sqrt{1-C^2\lambda_j^2}\,|0\rangle + C\lambda_j |1\rangle\right)$ **(D)** ✓ $\sum_j \beta_j |u_j\rangle |k_j\rangle \left(\sqrt{1-\frac{C^2}{\lambda_j^2}}\,|0\rangle + \frac{C}{\lambda_j}|1\rangle\right)$

Answer: (D) Detailed solution

Correct answer. (D) $\sum_j \beta_j |u_j\rangle |k_j\rangle \left(\sqrt{1-\frac{C^2}{\lambda_j^2}}\,|0\rangle + \frac{C}{\lambda_j}|1\rangle\right)$

Concepts required. State after eigenvalue inversion.

Relevant formulae. $R_Y(2\arcsin\frac{C}{\lambda})|0\rangle = \sqrt{1-\frac{C^2}{\lambda^2}}|0\rangle + \frac{C}{\lambda}|1\rangle$.

Step-by-step explanation. Write $|b\rangle = \sum_j \beta_j |u_j\rangle$. Phase estimation entangles each eigencomponent with its clock value: $\sum_j \beta_j |u_j\rangle |k_j\rangle |0\rangle$. The controlled rotation reads k_j and rotates the ancilla by $R_Y(2\arcsin(C/\lambda_j))$: $|0\rangle \to \sqrt{1-C^2/\lambda_j^2}\,|0\rangle + (C/\lambda_j)|1\rangle$. Uncomputation then resets the clock to $|0\rangle$, and the ancilla-$|1\rangle$ branch is $C\sum_j(\beta_j/\lambda_j)|u_j\rangle = C\,A^{-1}|b\rangle$. Option 2 multiplies by λ (that would apply A, not A^{-1}); option 3 is the final *unnormalised* target; option 4 has already uncomputed the clock.

Why the other options are incorrect.

(A) The clock is still entangled at this stage; it is reset only by uncomputation.

(B) That is the final target branch, after uncomputation and post-selection, without normalisation.

(C) Multiplying by λ would apply A, not A^{-1}.

Key takeaway. The ancilla's $|1\rangle$ amplitude carries C/λ_j.

Q.42 Single-correct MCQ · Qubit count

Using the notebook's layout (clock + system + one ancilla), an 8×8 Hermitian system with $n_c = 5$ clock qubits needs how many qubits?

(A) 14 **(B)** ✓ 9 **(C)** 13 **(D)** 8

Answer: (B) Detailed solution

Correct answer. (B) 9

Concepts required. Register sizes.

Relevant formulae. qubits $= n_c + \log_2 d + 1$.

Step-by-step explanation. System: $n_b = \log_2 8 = 3$ qubits (amplitude encoding of b). Total $n_c + n_b + 1 = 5 + 3 + 1 = 9$ — the notebook prints "Qubit requirement: n_b + n_c + 1". The default (4×4, $n_c = 4$) needs $4 + 2 + 1 = 7$. $8 + 5 + 1 = 14$ forgets that the system register is logarithmic in the dimension.

Why the other options are incorrect.

(A) 14 uses $d = 8$ system qubits instead of $\log_2 8 = 3$.

(C) 13 adds d and n_c without the logarithm.

(D) 8 is the dimension, not a qubit count.

Key takeaway. System qubits grow as $\log d$.

Q.43 Single-correct MCQ · Role of the constant C

The notebook sets $C = 2\pi/(2^{n_c}t)$, the smallest non-zero eigenvalue magnitude the clock can represent. Why must $C \le |\lambda_j|$ for every eigenvalue?

(A) ✓ The rotation amplitude C/λ_j must satisfy $|C/\lambda_j| \le 1$ so that $\arcsin(C/\lambda_j)$ exists

(B) Otherwise phase estimation needs more counting qubits than system qubits

(C) C must be the largest eigenvalue so that the success probability is 1

(D) Otherwise e^{iAt} would not be unitary

Answer: (A) Detailed solution

Correct answer. **(A)** The rotation amplitude C/λ_j must satisfy $|C/\lambda_j| \le 1$ so that $\arcsin(C/\lambda_j)$ exists

Concepts required. Role of the constant C.

Relevant formulae. $|C/\lambda_j| \le 1$; $P_{\text{success}} = \sum |\beta_j|^2 C^2/\lambda_j^2$.

Step-by-step explanation. The ancilla gets amplitude C/λ_j on $|1\rangle$; amplitudes must have modulus ≤ 1, so $C \le \min_j |\lambda_j|$. The success probability $\sum |\beta_j|^2 C^2/\lambda_j^2$ is largest when C is as large as allowed, $C = |\lambda|_{\min}$ — exactly what the `"auto"` time achieves (default: $C = 1 = \lambda_{\min}$). Unitarity of e^{iAt} needs only Hermitian A.

Why the other options are incorrect.

(B) C does not relate counting and system qubit numbers.

(C) $C = \lambda_{\max}$ would make $C/\lambda_{\min} > 1$, impossible.

(D) Unitarity of e^{iAt} depends only on Hermiticity.

Key takeaway. Choose C as large as possible: $C = |\lambda|_{\min}$.

Q.44 Single-correct MCQ · Rotation angle

With $C = 1$, the eigenvalue-inversion rotation for the clock value of $\lambda = 2$ is $R_Y(\theta)$ with

(A) $\theta = \frac{\pi}{2}$ **(B)** $\theta = \arcsin\frac{1}{2} = \frac{\pi}{6}$ **(C)** $\theta = 2\arcsin 2$ **(D)** ✓ $\theta = 2\arcsin\frac{1}{2} = \frac{\pi}{3}$

Answer: (D) Detailed solution

Correct answer. **(D)** $\theta = 2\arcsin\frac{1}{2} = \frac{\pi}{3}$

Concepts required. Rotation angle for the inversion.

Relevant formulae. $\theta = 2\arcsin(C/\lambda)$.

Step-by-step explanation. $\theta = 2\arcsin(C/\lambda) = 2\arcsin\frac{1}{2} = 2\cdot\frac{\pi}{6} = \frac{\pi}{3}$; then $R_Y(\theta)|0\rangle = \cos\frac{\pi}{6}|0\rangle + \sin\frac{\pi}{6}|1\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$, amplitude $\frac{1}{2} = 1/\lambda$ on $|1\rangle$. The factor 2 comes from the half-angle convention $R_Y(\theta) = e^{-i\theta Y/2}$; $\arcsin 2$ does not exist (why $C \le |\lambda|$).

Why the other options are incorrect.

(A) $\frac{\pi}{2}$ would give amplitude $\sin\frac{\pi}{4} \ne \frac{1}{2}$.

(B) $\frac{\pi}{6}$ forgets the half-angle factor 2.

(C) $\arcsin 2$ is undefined; C/λ not λ/C.

Key takeaway. Half-angle convention: $R_Y(\theta)$ rotates by $\theta/2$ in amplitude.

Q.45 Single-correct MCQ · Success probability (default)

For the default system ($\beta_j = \frac{1}{2}$ for the eigenvalues $1, 2, 3, 4$; $C = 1$), the ideal success probability $P_{\text{success}} = \sum_j |\beta_j|^2 C^2/\lambda_j^2$ is

(A) 1.4236 **(B)** 0.25 **(C)** ✓ 0.3559 **(D)** 0.5208

Answer: (C) Detailed solution

Correct answer. **(C)** 0.3559
Concepts required. Success probability.
Relevant formulae. $P = \sum_j |\beta_j|^2 C^2/\lambda_j^2 = C^2\|x\|^2/\|b\|^2$.
Step-by-step explanation. $P = \frac{1}{4}\left(1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16}\right) = \frac{1}{4} \cdot 1.423611 = 0.355903$ — the value printed by the notebook (state 0.355903, shots 0.3543). Equivalently $P = C^2\|A^{-1}b\|^2/\|b\|^2 = \|x\|^2 = 0.5966^2$. 1.4236 forgets the weights $|\beta_j|^2 = \frac{1}{4}$; 0.5208 is x_0.
Why the other options are incorrect.
(A) 1.4236 forgets the weights $\frac{1}{4}$.
(B) 0.25 keeps only the $\lambda = 1$ term.
(D) $0.5208 = \frac{25}{48}$ is the component x_0, not a probability.
Key takeaway. $P_{\text{success}} = \|x\|^2$ when $C = \|b\| = 1$.

Q.46 Matching lists · Register transformations

Match each stage of the notebook's HHL circuit (List-I) with what it does to the registers (List-II).

List-I		List-II	
(P)	amplitude initialisation	**(1)**	$\|u_j\rangle\|k_j\rangle\|0\rangle \to \|u_j\rangle\|k_j\rangle\left(\sqrt{1 - C^2/\lambda_j^2}\,|0\rangle + \frac{C}{\lambda_j}|1\rangle\right)$
(Q)	phase estimation	**(2)**	returns the clock register to $\|0\ldots0\rangle$ so that it is no longer entangled with the system
(R)	eigenvalue inversion	**(3)**	prepares $\|0\ldots0\rangle_{\text{clock}}\|b\rangle\|0\rangle_{\text{anc}}$
(S)	uncomputation	**(4)**	$\|b\rangle\|0\rangle \to \sum_j \beta_j\|u_j\rangle\|k_j\rangle$

(A) ✓ P→3, Q→4, R→1, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→4, Q→3, R→1, S→2 **(D)** P→3, Q→1, R→4, S→2

Answer: (A) Detailed solution

Correct answer. **(A)** P→3, Q→4, R→1, S→2
Concepts required. Register transformations of HHL.
Relevant formulae. $|0\rangle|b\rangle|0\rangle \to \sum \beta_j|u_j\rangle|k_j\rangle|0\rangle \to \cdots \to C|x\rangle|0\rangle|1\rangle + \ldots$
Step-by-step explanation.

- P: `kernel.qalloc(...)` loads $|0\rangle|b\rangle|0\rangle$ in one step (3).
- Q: Hadamards, controlled U^{2^j}, inverse QFT write the eigenvalue into the clock (4).
- R: the multi-controlled R_Y rotations (1).
- S: QFT, inverse controlled powers, Hadamards (2).

Why the other options are incorrect.
(B) R→2, S→1 swaps inversion and uncomputation.
(C) P→4, Q→3 swaps preparation and phase estimation.
(D) Q→1, R→4 swaps phase estimation and inversion.

Key takeaway. Prepare, estimate, invert, uncompute, post-select.

Q.47 Single-correct MCQ · Automatic evolution time

With `EVOLUTION_TIME = "auto"` the notebook chooses $t = 2\pi/(2^{n_c}|\lambda|_{\min})$ whenever $\kappa \le 2^{n_c-1}-1$. For $n_c = 4$ and eigenvalues $0.5, 1, 2, 3$, it picks

(A) $t = \pi/8$ and $C = 1$
(B) $t = 2\pi/3$ and $C = 3$
(C) $t = \pi/4$ and $C = 1$
(D) ✓ $t = \pi/4$ and $C = 0.5$

Answer: (D) Detailed solution

Correct answer. **(D)** $t = \pi/4$ and $C = 0.5$
Concepts required. The notebook's automatic evolution time.
Relevant formulae. $t = 2\pi/(2^{n_c}|\lambda|_{\min})$ if $\kappa \le 2^{n_c-1}-1$; $C = |\lambda|_{\min}$.
Step-by-step explanation. $\kappa = 3/0.5 = 6 \le 2^3 - 1 = 7$, so $t = 2\pi/(16 \cdot 0.5) = \pi/4$ and $C = 2\pi/(16t) = 0.5 = |\lambda|_{\min}$. The clock values $k = \lambda t \cdot 16/(2\pi) = 2\lambda$ are $1, 2, 4, 6$ — all exact and below the largest positive value 7. ($t = \pi/8$, $C = 1$ is right for the default with $\lambda_{\min} = 1$.)
Why the other options are incorrect.
(A) $t = \pi/8$, $C = 1$ belongs to $\lambda_{\min} = 1$.
(B) t is set by $\lambda_{\min}$, not $\lambda_{\max}$.
(C) $C = 1 > \lambda_{\min} = 0.5$ is not allowed.
Key takeaway. The automatic t maps $|\lambda|_{\min}$ to clock value 1.

Q.48 Single-correct MCQ · Why uncompute

Why does HHL uncompute phase estimation after the eigenvalue inversion instead of simply measuring the ancilla?

(A) Uncomputation doubles the success probability
(B) Uncomputation is needed to make e^{iAt} unitary
(C) It converts the amplitudes C/λ_j into probabilities
(D) ✓ Without it the clock stays entangled with the system, so the post-selected system register would be a mixture instead of $|x\rangle$ (unless the clock is also post-selected on a fixed value)

Answer: (D) Detailed solution

Correct answer. **(D)** Without it the clock stays entangled with the system, so the post-selected system register would be a mixture instead of $|x\rangle$ (unless the clock is also post-selected on a fixed value)
Concepts required. Why uncompute.
Relevant formulae. $\mathrm{tr}_{\text{clock}}\left|\sum \frac{\beta_j}{\lambda_j}|u_j\rangle|k_j\rangle\right|^2$ is a mixture.
Step-by-step explanation. The ancilla-$|1\rangle$ branch before uncomputation is $C\sum_j(\beta_j/\lambda_j)|u_j\rangle|k_j\rangle$: tracing out the clock leaves the incoherent mixture $\sum_j |\beta_j/\lambda_j|^2 |u_j\rangle\langle u_j|$, not the superposition $|x\rangle$. Reversing phase estimation maps $|k_j\rangle \to |0\rangle$ for every j, so the branch becomes $C|x\rangle \otimes |0\rangle$. The notebook post-selects ancilla $= 1$ *and* clock $= 0$. The success probability is unchanged by uncomputation.
Why the other options are incorrect.
(A) Uncomputation does not change the success probability.
(B) Unitarity of e^{iAt} needs only Hermitian A.
(C) Measurement, not uncomputation, converts amplitudes to probabilities.
Key takeaway. Disentangle ancillas before using a state.

Q.49 Assertion–Reason · Norm recovery

Assertion (A): From the simulated state vector, the notebook recovers the full solution x including its norm.

Reason (R): In the post-selected branch (clock = 0, ancilla = 1) the system amplitudes are $\frac{C}{\|b\|}x$, so $x = \frac{\|b\|}{C} \times$ (amplitudes).

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

Concepts required. Norm recovery from the state vector.

Relevant formulae. branch $= \frac{C}{\|b\|}x \Rightarrow x = \frac{\|b\|}{C} \times$ amplitudes.

Step-by-step explanation. **R is true:** the branch is $C\,A^{-1}\,|b\rangle = \frac{C}{\|b\|}A^{-1}b$ because $|b\rangle = b/\|b\|$. **A follows:** the code computes `x_quantum_full = amplitudes * b_norm / C_constant` and gets $x = (0.520833, 0.145833, 0.229167, 0.104167)$ with relative error 1.7×10^{-15}. (On hardware only $|x_i|^2/\|x\|^2$ is sampled.) Answer: (A).

Why the other options are incorrect.

(B) needs A and R both true but R not explaining A; R is exactly how the code does it.

(C) needs A true and R false; R is true.

(D) needs A false and R true; A is true.

Key takeaway. Simulators can read amplitudes; hardware cannot.

Q.50 Single-correct MCQ · Number of inversion rotations

The notebook's eigenvalue inversion applies one multi-controlled rotation for every non-zero clock value. With n_c clock qubits this is

(A) n_c rotations

(B) ✓ $2^{n_c} - 1$ rotations (15 for $n_c = 4$)

(C) one rotation per eigenvalue of A

(D) 2^{n_c} rotations, including $k = 0$

Answer: (B) Detailed solution

Correct answer. **(B)** $2^{n_c} - 1$ rotations (15 for $n_c = 4$)

Concepts required. Lookup-table eigenvalue inversion.

Relevant formulae. one rotation per $k = 1, \ldots, 2^{n_c} - 1$.

Step-by-step explanation. Clock values $k = 1, \ldots, 2^{n_c} - 1$ each get their own $R_Y(2\arcsin(C/\lambda_k))$; $k = 0$ would mean $\lambda = 0$ and is skipped. For $n_c = 4$: 15, matching `ccccry: 15`. The rotation count grows exponentially with the clock size — a lookup table, not an arithmetic $1/\lambda$ circuit; the matrix itself is not consulted.

Why the other options are incorrect.

(A) n_c rotations would need an arithmetic $1/\lambda$ circuit.

(C) The rotations do not depend on the eigenvalues of A but on all clock values.

(D) $k = 0$ ($\lambda = 0$) is skipped.

Key takeaway. Lookup tables cost 2^{n_c}; scalable versions compute $1/\lambda$.

Q.51 Single-correct MCQ · Negative clock values

With $n_c = 4$ and $t = \pi/8$, the notebook assigns to the clock value $k = 12$ the eigenvalue

(A) 12 **(B)** ✓ -4 **(C)** 4 **(D)** -12

Answer: (B) Detailed solution

Correct answer. (B) -4
Concepts required. Signed clock values.
Relevant formulae. $\lambda_k = 2\pi\,\mathrm{signed}(k)/(2^{n_c}t)$.
Step-by-step explanation. $12 \geq 2^{n_c-1} = 8$, so the signed value is $12 - 16 = -4$, and $\lambda_k = 2\pi \cdot (-4)/(16 \cdot \pi/8) = -4$. The rotation angle is $2\arcsin(C/\lambda) = 2\arcsin(-\frac{1}{4}) < 0$: the ancilla amplitude $-\frac{1}{4}$ carries the sign of $1/\lambda$. Without two's complement, negative eigenvalues would be misread as large positive ones.
Why the other options are incorrect.
(A) 12 ignores two's complement.
(C) 4 drops the sign: values ≥ 8 are negative in two's complement.
(D) -12 negates instead of subtracting 16.
Key takeaway. The sign of $1/\lambda$ is carried by the rotation angle.

Q.52 Statement I / II · Exact vs. rounded clock values

Statement I: If some $\lambda_j t\, 2^{n_c}/(2\pi)$ is not an integer, phase estimation spreads that component over several clock values and the fidelity of $|x\rangle$ drops below 1.

Statement II: The notebook's `"auto"` evolution time guarantees exact clock values for every matrix.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) ✓ Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

Correct answer. (C) Statement I is true, but Statement II is false
Concepts required. Exactness of the clock values.
Relevant formulae. exact iff $\lambda_j t\, 2^{n_c}/(2\pi) \in \mathbb{Z}$ for all j.
Step-by-step explanation. **I is true:** a non-integer clock value gives the leaky QPE distribution, and each leaked clock value applies the wrong $1/\lambda$ (with only 3 clock qubits the default reaches fidelity 0.953). **II is false:** `"auto"` only maps $|\lambda|_{\min}$ to clock value 1; other eigenvalues are exact only if they are integer multiples of $|\lambda|_{\min}$ (the notebook prints "(rounded -> error)" otherwise). Answer: (C).
Why the other options are incorrect.
(A) needs both statements true; II is false.
(B) needs both statements false; I is true.
(D) needs I false and II true; I is true and II is false.
Key takeaway. Read the notebook's "(exact)" / "(rounded)" labels.

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Q.53 Single-correct MCQ · 2×2 example: t, C

For the notebook's 2×2 example ($\lambda = \frac{2}{3}, \frac{4}{3}$) with $n_c = 4$ and `"auto"` time, the values of t, C and the clock values are

(A) $t = \frac{3\pi}{16}$, $C = \frac{4}{3}$, clock values 1 and 2
(B) $t = \frac{3\pi}{8}$, $C = \frac{2}{3}$, clock values 2 and 4
(C) $t = \frac{\pi}{8}$, $C = 1$, clock values $\frac{2}{3}$ and $\frac{4}{3}$
(D) ✓ $t = \frac{3\pi}{16}$, $C = \frac{2}{3}$, clock values 1 and 2

Answer: (D) Detailed solution

Correct answer. (D) $t = \frac{3\pi}{16}$, $C = \frac{2}{3}$, clock values 1 and 2
Concepts required. Automatic t for the 2×2 example.
Relevant formulae. $t = 2\pi/(2^{n_c}\lambda_{\min})$, $C = \lambda_{\min}$, $k = \lambda/\lambda_{\min}$.
Step-by-step explanation. $\kappa = 2 \le 7$: $t = 2\pi/(16 \cdot \frac{2}{3}) = \frac{3\pi}{16}$, $C = 2\pi/(16t) = \frac{2}{3}$. Clock values $k = \lambda t \cdot 16/(2\pi) = \lambda \cdot \frac{3}{2}$: 1 and 2 — exact, so HHL is exact (simulated fidelity 1, $x = (1.125, 0.375)$, $P_{\text{success}} = 0.625$). $C = \frac{4}{3}$ would exceed $\lambda_{\min}$.
Why the other options are incorrect.
(A) $C = \frac{4}{3} > \lambda_{\min}$ is not allowed.
(B) $t = \frac{3\pi}{8}$ doubles the time and the clock values.
(C) $t = \pi/8$ belongs to $\lambda_{\min} = 1$; clock values must be integers.
Key takeaway. Exact mapping ⇒ exact HHL (fidelity 1).

Q.54 Single-correct MCQ · Post-selection condition

In the sampling read-out the notebook keeps a shot only if

(A) the clock bits encode an eigenvalue
(B) the ancilla bit is 0
(C) ✓ all clock bits are 0 and the ancilla bit is 1
(D) the system bits are all 0

Answer: (C) Detailed solution

Correct answer. (C) all clock bits are 0 and the ancilla bit is 1
Concepts required. The notebook's post-selection.
Relevant formulae. accept iff clock $= 0 \ldots 0$ and ancilla $= 1$.
Step-by-step explanation. Code: `if bits[:n_c] == "0" * n_c and bits[-1] == "1"`. Ancilla = 1 selects the branch carrying C/λ; clock = 0 selects the correctly uncomputed part (in the exact case this is automatic). The system bits of the accepted shots are then histogrammed to estimate $|x_i|^2/\|x\|^2$.
Why the other options are incorrect.
(A) The clock must be back in $|0\rangle$ after uncomputation.
(B) Ancilla 0 is the failure branch.
(D) System bits are the data, not the acceptance condition.
Key takeaway. Accepted shots sample $|x_i|^2/\|x\|^2$.

Q.55 SINGLE-CORRECT MCQ · Shot statistics of P_{success}

The ideal success probability is $p = 0.3559$ and the notebook uses 20 000 shots. The standard error of the measured fraction of accepted shots is about

(A) ✓ 0.0034 **(B)** 0.0001 **(C)** 0.0119 **(D)** 0.048

Answer: (A) Detailed solution

Correct answer. **(A)** 0.0034
Concepts required. Binomial standard error.
Relevant formulae. $\text{SE} = \sqrt{p(1-p)/S}$.
Step-by-step explanation. Binomial: $\sqrt{p(1-p)/S} = \sqrt{0.3559 \times 0.6441/20000} = 0.0034$. The recorded fraction 0.3543 differs from 0.3559 by 0.0016, about half a standard error — consistent. $0.0119 = 1/\sqrt{7086}$ is the notebook's per-component scale for the *accepted* shots.
Why the other options are incorrect.
(B) 0.0001 forgets the square root.
(C) 0.0119 is $1/\sqrt{7086}$, a per-component scale.
(D) 0.048 uses $\sqrt{p(1-p)/100}$.
Key takeaway. The recorded 0.3543 is within half a standard error of 0.3559.

Q.56 SINGLE-CORRECT MCQ · What sampling reveals

On hardware, the post-selected system bitstrings of HHL directly estimate

(A) $\|x\|$ only
(B) ✓ $|x_i|^2/\|x\|^2$ only — signs/phases and the norm need further measurements
(C) $x_i/\|x\|$ including its sign
(D) every component x_i including its sign

Answer: (B) Detailed solution

Correct answer. **(B)** $|x_i|^2/\|x\|^2$ only — signs/phases and the norm need further measurements
Concepts required. Read-out limits of HHL.
Relevant formulae. $P(i) = |x_i|^2/\|x\|^2$; tomography cost $\sim d/\varepsilon^2$.
Step-by-step explanation. Computational-basis measurement gives probabilities $|\langle i|x\rangle|^2$. Signs need interference with other bases; the norm needs P_{success} (with $\|b\|$ and C). The notebook shows this: sampled $|x_i|^2$ $(0.7691, 0.0587, 0.1420, 0.0302)$ vs. exact $(0.7622, 0.0598, 0.1476, 0.0305)$, magnitudes only. Reading all d components to precision ε costs about d/ε^2 shots.
Why the other options are incorrect.
(A) The norm comes from P_{success}, not the histogram.
(C) Same: only magnitudes are sampled.
(D) Signs are invisible in computational-basis probabilities.
Key takeaway. HHL is for properties of x, not for printing it.

Q.57 SINGLE-CORRECT MCQ · Multi-qubit gate count

The recorded circuit statistics of the default HHL circuit are `h: 16, ccustom_unitary_constant: 8, swap: 4, cr1: 12, x: 56, ccccry: 15`. How many multi-qubit gates is that?

(A) 111 **(B)** ✓ 39 **(C)** 23 **(D)** 95

Answer: (B) Detailed solution

Correct answer. (B) 39
Concepts required. Counting multi-qubit gates.
Relevant formulae. $8 + 4 + 12 + 15 = 39$.
Step-by-step explanation. Multi-qubit: 8 controlled $U^{\pm 2^j}$ + 4 swaps + 12 `cr1` + 15 multi-controlled $R_Y = 39$, the recorded "multi-qubit gates: 39". Single-qubit: $16 + 56 = 72$; total 111. The 8 custom gates are 4 controlled U^{2^j} in phase estimation and 4 inverses in the uncomputation.
Why the other options are incorrect.
(A) 111 is the total gate count.
(C) $23 = 8 + 15$ counts only the controlled U and the rotations; it omits the 4 swaps and 12 `cr1`.
(D) 95 counts X gates as multi-qubit.
Key takeaway. Most gates are cheap X polarity gates; the expensive ones are the controlled U and $C^4 R_Y$.

Q.58 MATCHING LISTS · HHL ↔ code

Match each HHL ingredient (List-I) with the code that implements it in `HHL.ipynb` (List-II).

List-I		List-II	
(P)	preparation of $\|0\rangle\|b\rangle\|0\rangle$	**(1)**	`kernel.cry(theta, clock, ancilla)` between X gates
(Q)	controlled powers U^{2^j}	**(2)**	`postselected_amplitudes(state_vector)`
(R)	eigenvalue inversion	**(3)**	`kernel.qalloc(bit_reverse_order(initial, n_tot))`
(S)	extraction of the post-selected amplitudes	**(4)**	`cudaq.register_operation(name, matrix_power(U, 2**j))`

(A) P→4, Q→3, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2
(C) ✓ P→3, Q→4, R→1, S→2 **(D)** P→3, Q→4, R→2, S→1

Answer: (C) Detailed solution

Correct answer. (C) P→3, Q→4, R→1, S→2
Concepts required. Mapping HHL to the notebook's code.
Relevant formulae. `qalloc` → prep; `register_operation` → U^{2^j}; `cry` → inversion.
Step-by-step explanation.

- P: amplitude initialisation of the whole register (qubit order converted for CUDA-Q) (3).
- Q: each power is registered as a custom unitary and applied controlled by one clock qubit (4).
- R: the multi-controlled R_Y with polarity X gates (1).
- S: the helper picks the system amplitudes with clock = 0 and ancilla = 1 (2).

Why the other options are incorrect.
(A) P→4, Q→3 swaps preparation and controlled powers.

(B) Q→1, R→4 swaps controlled powers and inversion.

(D) R→2, S→1 swaps inversion and extraction.

Key takeaway. Every HHL stage has one recognisable line of code.

Paragraph for Questions 59 and 60

HHL in the notebook's convention is applied to $A = \mathrm{diag}(1, 2)$ and $b = \frac{1}{\sqrt{2}}(1, 1)$ with $n_c = 3$ clock qubits and $t = \pi/4$, so $U = e^{iAt}$, the clock value of an eigenvalue λ is $k = \lambda t\, 2^{n_c}/(2\pi) = \lambda$, and $C = 2\pi/(2^{n_c} t) = 1$. The eigenvalue inversion applies $R_Y(2\arcsin(C/\lambda_k))$ to the ancilla for each clock value; the system register has one qubit.

Q.59 PARAGRAPH BASED · Passage: state after QPE

After phase estimation the state of system ⊗ clock (ancilla still $|0\rangle$) is

(A) $\frac{1}{\sqrt{2}}(\,|0\rangle\,|001\rangle + |1\rangle\,|011\rangle\,)$

(B) $\frac{1}{\sqrt{2}}(\,|0\rangle\,|100\rangle + |1\rangle\,|010\rangle\,)$

(C) ✓ $\frac{1}{\sqrt{2}}(\,|0\rangle\,|001\rangle + |1\rangle\,|010\rangle\,)$

(D) $\frac{1}{\sqrt{2}}(\,|0\rangle + |1\rangle\,)\,|001\rangle$

Answer: (C) Detailed solution

Correct answer. **(C)** $\frac{1}{\sqrt{2}}(\,|0\rangle\,|001\rangle + |1\rangle\,|010\rangle\,)$

Concepts required. Phase estimation on a diagonal matrix.

Relevant formulae. $|b\rangle = \sum \beta_j\,|u_j\rangle \to \sum \beta_j\,|u_j\rangle\,|k_j\rangle$.

Step-by-step explanation. A is diagonal, so $|0\rangle$ and $|1\rangle$ are eigenvectors with $\lambda = 1, 2$, and $\beta = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Clock values $k = \lambda$: $1 = |001\rangle$ and $2 = |010\rangle$ (most significant bit first), both exact. Result: $\frac{1}{\sqrt{2}}(|0\rangle\,|001\rangle + |1\rangle\,|010\rangle)$ — system and clock entangled. $|100\rangle$ is 1 written in reverse bit order.

Why the other options are incorrect.

(A) $|011\rangle = 3$ is not the clock value of $\lambda = 2$.

(B) $|100\rangle$ is 1 written with the least significant bit first.

(D) A product state would need $|b\rangle$ to be an eigenvector.

Key takeaway. Different eigenvalues → different clock values → entanglement.

Q.60 PARAGRAPH BASED · Passage: P_{success} and $|x\rangle$

The success probability and the normalised output state $|x\rangle$ are

(A) ✓ 0.625 and $\frac{1}{\sqrt{5}}(2, 1)$

(B) 0.625 and $\frac{1}{\sqrt{2}}(1, 1)$

(C) 0.75 and $\frac{1}{\sqrt{5}}(2, 1)$

(D) 1 and $\frac{1}{\sqrt{5}}(1, 2)$

Answer: (A) Detailed solution

Correct answer. **(A)** 0.625 and $\frac{1}{\sqrt{5}}(2, 1)$

Concepts required. Post-selected branch and normalisation.

Relevant formulae. $P = \sum |\beta_j|^2 C^2/\lambda_j^2$; $|x\rangle \propto \sum \beta_j \lambda_j^{-1}\,|u_j\rangle$.

Step-by-step explanation. Ancilla amplitudes $C/\lambda = 1$ and $\frac{1}{2}$: post-selected branch $\frac{1}{\sqrt{2}}(1 \cdot |0\rangle + \frac{1}{2}\,|1\rangle)$, probability $\frac{1}{2}(1 + \frac{1}{4}) = 0.625$. Normalising: $(1, \frac{1}{2})/\sqrt{1.25} = \frac{1}{\sqrt{5}}(2, 1)$, proportional to $A^{-1}b = \frac{1}{\sqrt{2}}(1, \frac{1}{2})$ ✓. (Full circuit simulation: $P = 0.625$, $|x\rangle = (0.8944, 0.4472)$.) $(1, 2)/\sqrt{5}$ would be $A\,|b\rangle$ normalised.

Why the other options are incorrect.

(B) $\frac{1}{\sqrt{2}}(1, 1)$ is $|b\rangle$, not $|x\rangle$.

(C) 0.75 averages the amplitudes instead of their squares.

(D) $(1, 2)/\sqrt{5}$ is $A\,|b\rangle$ normalised.

Key takeaway. Divide by λ, then normalise.

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SECTION D · Advanced HHL and Implementation Q.61–80

Q.61 Single-correct MCQ · HHL complexity

The original HHL analysis (sparse $d \times d$ matrix with sparsity s, condition number κ, precision ε) gives a running time of

(A) $O(\log d)$, independent of κ and ε
(B) $O(d\, s\sqrt{\kappa}\log(1/\varepsilon))$
(C) $O(d^3)$
(D) ✓ $\tilde{O}(\log(d)\, s^2\kappa^2/\varepsilon)$

Answer: (D) Detailed solution

Correct answer. (D) $\tilde{O}(\log(d)\, s^2\kappa^2/\varepsilon)$
Concepts required. HHL complexity.
Relevant formulae. $\tilde{O}(\log(d)s^2\kappa^2/\varepsilon)$.
Step-by-step explanation. Harrow, Hassidim and Lloyd (2009): $\tilde{O}(\log(d)s^2\kappa^2/\varepsilon)$ — logarithmic in d thanks to amplitude encoding and sparse Hamiltonian simulation, but polynomial in κ and $1/\varepsilon$. $O(d^3)$ is Gaussian elimination, $O(ds\sqrt{\kappa}\log(1/\varepsilon))$ is conjugate gradient. The exponential advantage in d only holds if κ and s stay small, $|b\rangle$ is easy to prepare and only a property of $|x\rangle$ is needed.
Why the other options are incorrect.
(A) HHL depends polynomially on κ and $1/\varepsilon$.
(B) $O(ds\sqrt{\kappa}\log(1/\varepsilon))$ is conjugate gradient.
(C) $O(d^3)$ is Gaussian elimination.
Key takeaway. Exponential in d only; polynomial in κ, s, $1/\varepsilon$.

Q.62 Single-correct MCQ · Amplitude amplification

HHL succeeds (ancilla = 1) with probability $p = 0.01$. On average, how many repetitions are needed with plain repetition, and with amplitude amplification?

(A) ✓ about 100 and about 10
(B) about 100 and about 1
(C) about 1 and about 10
(D) about 10 and about 100

Answer: (A) Detailed solution

Correct answer. (A) about 100 and about 10
Concepts required. Amplitude amplification.
Relevant formulae. repetitions $1/p$ vs. $O(1/\sqrt{p})$.
Step-by-step explanation. Repeat-until-success needs $1/p = 100$ trials on average. Amplitude amplification rotates the success amplitude $\sqrt{p}$ towards 1 in $O(1/\sqrt{p}) = O(10)$ iterations. Since $p \sim 1/\kappa^2$ in the worst case, amplification turns HHL's κ^2 repetition cost into κ — part of the original HHL analysis. The notebook simply post-selects (here $p = 0.356$).
Why the other options are incorrect.
(B) Amplification needs $\sim 1/\sqrt{p} = 10$ rounds, not 1.
(C) Plain repetition needs $1/p = 100$, not 1.
(D) Reversed: plain repetition needs about $1/p = 100$ trials, amplification about $1/\sqrt{p} = 10$.
Key takeaway. Amplification turns the κ^2 repetition cost into κ.

Q.63 Single-correct MCQ · Read-out cost

To estimate all $d = 1024$ probabilities $|x_i|^2$ of $|x\rangle$ to absolute precision $\varepsilon = 0.01$ by sampling, roughly how many post-selected shots are needed?

(A) about 10^{14} **(B)** about 10 **(C)** ✓ about 10^7 **(D)** about 10^3

Answer: (C) Detailed solution

Correct answer. (C) about 10^7
Concepts required. Tomography cost.
Relevant formulae. $\sim d/\varepsilon^2$ shots.
Step-by-step explanation. Each estimate needs $\sim 1/\varepsilon^2 = 10^4$ samples' worth of precision, and d components share the budget: the usual scaling is $d/\varepsilon^2 = 1024 \times 10^4 \approx 10^7$ shots. This is linear in d and kills the exponential speed-up — the reason HHL is useful for *properties* of x (e.g. $x^\dagger Mx$), not for printing x. The notebook's text says the same ("about d/ϵ^2 shots").
Why the other options are incorrect.
(A) 10^{14} squares d as well.
(B) 10 is far too few for 1024 components.
(D) 10^3 ignores the $1/\varepsilon^2$ factor.
Key takeaway. Reading all of x cancels the exponential speed-up.

Q.64 Matching lists · Clock size for κ

For exact clock values the notebook needs $\kappa \le 2^{n_c-1} - 1$. Match each condition number (List-I) with the smallest suitable number of clock qubits n_c (List-II).

List-I		List-II	
(P)	$\kappa = 3$	**(1)**	6
(Q)	$\kappa = 7$	**(2)**	3
(R)	$\kappa = 20$	**(3)**	7
(S)	$\kappa = 40$	**(4)**	4

(A) P→2, Q→4, R→3, S→1 **(B)** P→2, Q→1, R→4, S→3
(C) P→4, Q→2, R→1, S→3 **(D)** ✓ P→2, Q→4, R→1, S→3

Answer: (D) Detailed solution

Correct answer. (D) P→2, Q→4, R→1, S→3
Concepts required. Clock size for a given condition number.
Relevant formulae. need $2^{n_c-1} - 1 \ge \kappa$.
Step-by-step explanation. Largest positive signed clock value $2^{n_c-1} - 1$: $n_c = 3 \to 3$, $4 \to 7$, $5 \to 15$, $6 \to 31$, $7 \to 63$.

- P: $\kappa = 3 \le 3$: $n_c = 3$ (2). Q: $\kappa = 7 \le 7$: $n_c = 4$ (4).
- R: $\kappa = 20$: $15 < 20 \le 31$: $n_c = 6$ (1) — the notebook pairs `RANDOM_CONDITION_NUMBER = 20` with `NUM_CLOCK_QUBITS = 6`.
- S: $\kappa = 40$: $31 < 40 \le 63$: $n_c = 7$ (3).

One clock qubit is the sign bit, hence 2^{n_c-1} rather than 2^{n_c}.
Why the other options are incorrect.
(A) R→3, S→1 swaps $\kappa = 20$ ($n_c = 6$) and $\kappa = 40$ ($n_c = 7$).
(B) Q→1, R→4: $n_c = 4$ allows only $\kappa \le 7$, not 20.

(C) P→4, Q→2 swaps: $\kappa = 3$ already fits with $n_c = 3$, and $\kappa = 7$ needs $n_c = 4$.
Key takeaway. One clock qubit is the sign bit.

Q.65 Single-correct MCQ · Exponential rotation count

With `NUM_CLOCK_QUBITS = 8`, how many multi-controlled eigenvalue-inversion rotations does the notebook's circuit contain, and how many qubits does the default 4×4 system then use?

(A) ✓ 255 rotations, 11 qubits
(B) 255 rotations, 14 qubits
(C) 256 rotations, 12 qubits
(D) 8 rotations, 11 qubits

Answer: (A) Detailed solution

Correct answer. **(A)** 255 rotations, 11 qubits
Concepts required. Growth of the lookup-table inversion.
Relevant formulae. rotations $2^{n_c} - 1$; qubits $n_c + n_b + 1$.
Step-by-step explanation. One rotation per non-zero clock value: $2^8 - 1 = 255$, each controlled by all 8 clock qubits. Qubits: $8 + 2 + 1 = 11$ (the notebook's table: "11 qubits in total"). The lookup-table inversion is exponential in n_c; scalable implementations compute $1/\lambda$ arithmetically or use polynomial approximations (QSVT).
Why the other options are incorrect.
(B) 14 qubits counts the system as $d = 4$ qubits.
(C) $k = 0$ is skipped, and the ancilla count is 1.
(D) 8 rotations would need arithmetic inversion.
Key takeaway. Exponential gate growth in n_c limits precision.

Q.66 Single-correct MCQ · Embedding in the notebook

With `MATRIX_A = [[1, 2], [0, 1]]` (not Hermitian) and $n_c = 4$, the notebook's circuit uses how many qubits, and where is x in the solution of the embedded system?

(A) 7 qubits; x is the first half of the solution
(B) ✓ 7 qubits; x is the second half of the solution
(C) 8 qubits; x is the second half
(D) 6 qubits; x is the whole solution

Answer: (B) Detailed solution

Correct answer. **(B)** 7 qubits; x is the second half of the solution
Concepts required. Hermitian embedding in the notebook.
Relevant formulae. $\left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} y \\ x \end{smallmatrix}\right) = \left(\begin{smallmatrix} b \\ 0 \end{smallmatrix}\right)$.
Step-by-step explanation. Embedding $\left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} y \\ x \end{smallmatrix}\right) = \left(\begin{smallmatrix} b \\ 0 \end{smallmatrix}\right)$ gives $Ax = b$ (and $A^\dagger y = 0$, so $y = 0$). Dimension $4 \Rightarrow n_b = 2$; total $4 + 2 + 1 = 7$. The code sets `solution_slice = slice(d0, 2 * d0)`: the second half. The embedding costs one extra system qubit and its κ is that of the singular values (5.83 here).
Why the other options are incorrect.
(A) x is the second block (`slice(d0, 2*d0)`).
(C) 8 qubits would need 3 system qubits.
(D) The dimension doubles to 4, so 2 system qubits are needed.
Key takeaway. Embedding costs one qubit and changes κ to the singular-value ratio.

Q.67 Single-correct MCQ · Padding and κ

A 3×3 Hermitian matrix with eigenvalues $5, 10, 20$ is padded by the notebook to 4×4 with an identity block. What happens to κ?

(A) ✓ it grows from 4 to 20, because the padding adds the eigenvalue 1

(B) it stays 4

(C) it becomes 5

(D) it becomes 1

Answer: (A) Detailed solution

Correct answer. **(A)** it grows from 4 to 20, because the padding adds the eigenvalue 1
Concepts required. Padding with an identity block.
Relevant formulae. padded spectrum $= \{5, 10, 20\} \cup \{1\}$.
Step-by-step explanation. Before padding $\kappa = 20/5 = 4$. The identity block adds eigenvalue 1 (with b padded by zeros, its eigenvector carries no weight of $|b\rangle$), but the clock and C must still cover $\lambda = 1$ under the `"auto"` rule: $\kappa = 20/1 = 20$. A padding value inside the spectrum (e.g. 5–20) would avoid this; the notebook's identity padding is simple but can worsen the conditioning that sets the clock size.
Why the other options are incorrect.

(B) Padding adds the eigenvalue 1, which enters κ.

(C) 5 has no derivation.

(D) $\kappa = 1$ would need all eigenvalues equal.

Key takeaway. Choose padding values inside the spectrum to keep κ.

Q.68 Single-correct MCQ · Dequantisation

"Dequantised" algorithms (Tang, 2018 and later) showed that for some quantum linear-algebra tasks

(A) HHL can run without phase estimation

(B) the condition number no longer affects quantum algorithms

(C) classical computers can prepare $|x\rangle$ on a quantum computer

(D) ✓ classical randomized algorithms with comparable sampling access achieve similar poly-logarithmic dependence on d for low-rank matrices

Answer: (D) Detailed solution

Correct answer. **(D)** classical randomized algorithms with comparable sampling access achieve similar poly-logarithmic dependence on d for low-rank matrices
Concepts required. Dequantisation.
Relevant formulae. classical cost $\mathrm{poly}(\mathrm{rank}, \kappa, 1/\varepsilon) \cdot \mathrm{polylog}(d)$ with sample access.
Step-by-step explanation. Given "sample and query" access to the data (analogous to the qRAM assumed by quantum algorithms), classical sampling methods solve low-rank problems (recommendation systems, low-rank linear systems) in time polylogarithmic in d, polynomial in the rank and κ. Exponential quantum advantage therefore needs sparse, full-rank, well-conditioned structure and efficient state preparation — not just "a big matrix".
Why the other options are incorrect.

(A) Dequantisation is about classical algorithms, not about removing phase estimation.

(B) κ still matters for both.

(C) Classical computers do not prepare quantum states.

Key takeaway. Quantum advantage needs more than a large matrix.

Q.69 Assertion–Reason · Conditional speed-up

Assertion (A): HHL gives an exponential speed-up over classical algorithms for every linear system $Ax = b$.

Reason (R): For s-sparse, well-conditioned A with efficiently preparable $|b\rangle$, HHL prepares $|x\rangle$ in time polynomial in $\log d$, s, κ and $1/\varepsilon$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) ✓ A is false, but R is true

Answer: (D) Detailed solution

Correct answer. **(D)** A is false, but R is true
Concepts required. Conditions for HHL's speed-up.
Relevant formulae. poly$(\log d, s, \kappa, 1/\varepsilon)$ under assumptions.
Step-by-step explanation. **R is true** (the HHL theorem, under its assumptions). **A is false:** dense or ill-conditioned matrices ($\kappa \sim d$), expensive state preparation, the need to read all of x (d/ε^2 shots), and dequantised classical algorithms each remove the advantage. HHL's speed-up is conditional. Answer: (D).
Why the other options are incorrect.
(A) needs A true, R true *and* R explaining A; A is false.
(B) needs A and R both true but R not explaining A; A is false.
(C) needs A true and R false; A is false and R is true.
Key takeaway. Every assumption matters: sparsity, κ, state preparation, read-out.

Q.70 Single-correct MCQ · Worst-case success probability

Take $C = \lambda_{\min}$ and a matrix with $\kappa = 20$. If $|b\rangle$ is the eigenvector of $\lambda_{\max}$, the success probability is

(A) 0.5 **(B)** ✓ $1/\kappa^2 = 0.0025$ **(C)** 1 **(D)** $1/\kappa = 0.05$

Answer: (B) Detailed solution

Correct answer. **(B)** $1/\kappa^2 = 0.0025$
Concepts required. Extremes of the success probability.
Relevant formulae. $P = \sum |\beta_j|^2 \lambda_{\min}^2/\lambda_j^2 \in [1/\kappa^2, 1]$.
Step-by-step explanation. $P = \sum_j |\beta_j|^2 C^2/\lambda_j^2$ with all weight on $\lambda_{\max}$: $P = (\lambda_{\min}/\lambda_{\max})^2 = 1/\kappa^2 = 1/400$. If $|b\rangle$ were the eigenvector of $\lambda_{\min}$, $P = 1$. In general $1/\kappa^2 \le P \le 1$; the notebook's default ($\kappa = 4$, equal weights) gives 0.356.
Why the other options are incorrect.
(A) 0.5 has no derivation.
(C) 1 is the opposite extreme ($|b\rangle$ on $\lambda_{\min}$).
(D) $1/\kappa$ forgets to square.
Key takeaway. Worst-case success probability $1/\kappa^2$.

Q.71 SINGLE-CORRECT MCQ · Aliasing

With $n_c = 4$ and $t = \pi/8$ (clock value $k = \lambda$), how is an eigenvalue $\lambda = 9$ treated?

(A) Phase estimation fails with an error

(B) It is read as 7, the largest positive value

(C) ✓ It aliases: clock value 9 is read as the signed value -7, so the wrong (negative) $1/\lambda$ is applied

(D) It is read correctly as 9

Answer: (C) Detailed solution

Correct answer. (C) It aliases: clock value 9 is read as the signed value -7, so the wrong (negative) $1/\lambda$ is applied

Concepts required. Phase aliasing.

Relevant formulae. $k \equiv \lambda t\, 2^{n_c}/(2\pi) \pmod{2^{n_c}}$, signed.

Step-by-step explanation. $e^{i\lambda t}$ only knows λt modulo 2π: $k = 9 \pmod{16}$, and $9 \geq 8$ is interpreted as $9 - 16 = -7$. The inversion applies amplitude $C/(-7)$ instead of $C/9$ — wrong magnitude and sign. The notebook warns "some $|k_j|$ exceed the representable range -> eigenvalues alias; reduce EVOLUTION_TIME" (it requires $|\lambda|t < \pi$, here $|\lambda| < 8$).

Why the other options are incorrect.

(A) Phase estimation always returns some outcome; there is no error signal.

(B) Clock values wrap around, they are not clipped to 7.

(D) 9 is outside the signed range $[-8, 7]$.

Key takeaway. Keep $|\lambda|t < \pi$.

Q.72 MATCHING LISTS · Quantum linear-system algorithms

Match each quantum linear-system algorithm (List-I) with its key idea (List-II).

List-I		List-II	
(P)	HHL (2009)	**(1)**	implement a polynomial approximation of $1/x$ on the singular values of a block-encoded A
(Q)	Ambainis (2012), variable-time amplitude amplification	**(2)**	phase estimation + controlled rotation C/λ + post-selection; cost κ^2/ε
(R)	Childs–Kothari–Somma (2017)	**(3)**	amplify different eigenvalue branches at different times: κ dependence close to linear ($\kappa \log^3 \kappa$)
(S)	QSVT-based solvers	**(4)**	linear combination of unitaries (Fourier/Chebyshev series of $1/x$): poly $\log(1/\varepsilon)$ precision dependence

(A) P→2, Q→4, R→3, S→1

(B) P→3, Q→2, R→4, S→1

(C) ✓ P→2, Q→3, R→4, S→1

(D) P→2, Q→3, R→1, S→4

Answer: (C) Detailed solution

Correct answer. (C) P→2, Q→3, R→4, S→1

Concepts required. Quantum linear-system algorithms.

Relevant formulae. HHL κ^2/ε; VTAA κ; CKS polylog$(1/\varepsilon)$; QSVT polynomial of $1/x$.

Step-by-step explanation.

- P: HHL — the scheme implemented in the notebook (2).
- Q: variable-time amplitude amplification improves the κ dependence (3).

- R: Childs–Kothari–Somma replace phase estimation by LCU, improving $1/\varepsilon$ to $\operatorname{poly}\log(1/\varepsilon)$ (4).
- S: quantum singular value transformation unifies these with polynomial approximations (1). (Adiabatic/discrete-adiabatic methods reach optimal $\kappa \log(1/\varepsilon)$.)

Why the other options are incorrect.

(A) Q→4, R→3 swaps VTAA and CKS.

(B) P→3, Q→2 swaps HHL and VTAA.

(D) R→1, S→4 swaps CKS and QSVT.

Key takeaway. Newer algorithms improve the κ and ε dependence.

Q.73 SINGLE-CORRECT MCQ · Density matrix and global phase

With `SIMULATION_METHOD = "density_matrix"` the notebook multiplies its recovered x by a phase factor before comparing with NumPy. Why?

(A) ✓ A density matrix $|\psi\rangle\langle\psi|$ carries no global phase, so the vector extracted from it has an arbitrary phase that must be aligned

(B) The density-matrix simulator uses fp32 precision

(C) The density-matrix simulator adds noise that rotates the phase

(D) Post-selection on a density matrix flips the sign of every component

Answer: (A) Detailed solution

Correct answer. **(A)** A density matrix $|\psi\rangle\langle\psi|$ carries no global phase, so the vector extracted from it has an arbitrary phase that must be aligned

Concepts required. Global phase and density matrices.

Relevant formulae. $\rho = |\psi\rangle\langle\psi|$ invariant under $|\psi\rangle \to e^{i\alpha}|\psi\rangle$.

Step-by-step explanation. $\rho = |\psi\rangle\langle\psi|$ is invariant under $|\psi\rangle \to e^{i\alpha}|\psi\rangle$. The notebook takes the dominant eigenvector of ρ (`evecs[:, -1]`), whose phase is arbitrary, and aligns it with `overlap / abs(overlap)` — printing "a density matrix carries no global phase". No noise model is used in the HHL notebook, so the density-matrix result is still exact.

Why the other options are incorrect.

(B) The density-matrix target is fp64.

(C) No noise model is used in the HHL notebook.

(D) Post-selection does not flip signs.

Key takeaway. Compare states up to a global phase.

Q.74 STATEMENT I / II · Custom gates; hardware refused

Statement I: `HHL.ipynb` refuses hardware backends, because exact matrix exponentials as custom gates and amplitude initialisation cannot run on hardware targets.

Statement II: The controlled powers U^{2^j} are registered as custom unitary matrices with `cudaq.register_operation`.

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

Correct answer. **(A)** Both Statement I and Statement II are true
Concepts required. The notebook's custom gates and backends.
Relevant formulae. `register_operation`; hardware rejected with `ValueError`.
Step-by-step explanation. **I is true:** `if backend_cfg["device"] == "hardware": raise ValueError("This HHL implementation uses exact matrix exponentials as custom gates and amplitude encoding, which hardware targets cannot execute...").` **II is true:** `cudaq.register_operation(name, np.linalg.matrix_power(U, 2 ** j))`. On hardware, e^{iAt} would need a gate-level Hamiltonian simulation (e.g. Trotterised Pauli rotations), which the notebook does not implement. Answer: (A).
Why the other options are incorrect.
(B) needs both statements false; both are true.
(C) needs I true and II false; II is true.
(D) needs I false and II true; I is true.
Key takeaway. Simulator-only implementation; hardware needs Hamiltonian simulation.

Q.75 SINGLE-CORRECT MCQ · Too few clock qubits

Run with only `NUM_CLOCK_QUBITS = 3` on the default system ($\kappa = 4 > 3$ levels), the notebook rescales t so that $\lambda_{\max} = 4$ maps to clock value 3. The ideal clock values of $\lambda = 1, 2, 3, 4$ and the effect are

(A) $1, 2, 3, 4$ — still exact
(B) ✓ $0.75, 1.5, 2.25, 3$ — non-integer values are rounded by phase estimation, so the fidelity drops below 1
(C) $3, 6, 9, 12$ — all eigenvalues alias
(D) $0.75, 1.5, 2.25, 3$ — the fidelity stays exactly 1

Answer: (B) Detailed solution

Correct answer. **(B)** $0.75, 1.5, 2.25, 3$ — non-integer values are rounded by phase estimation, so the fidelity drops below 1
Concepts required. Rescaled time with too few clock qubits.
Relevant formulae. $t = 2\pi k_{\max}/(2^{n_c}\lambda_{\max})$, $k = 0.75\lambda$.
Step-by-step explanation. $t = 2\pi \cdot 3/(8 \cdot 4) = 3\pi/16$, so $k = \lambda t\, 2^3/(2\pi) = 0.75\lambda$: $0.75, 1.5, 2.25, 3$. Only $\lambda = 4$ is exact; the others leak over neighbouring clock values and are inverted with wrong $1/\lambda$. Simulation of the notebook's circuit (CUDA-Q) gives a fidelity of about 0.953 instead of 1 — the notebook's "Things to try next" predicts exactly this drop.
Why the other options are incorrect.
(A) With $n_c = 3$, $\lambda = 4$ cannot map to 4.
(C) Clock values $3, 6, 9, 12$ would need a time 4 times larger.
(D) Non-integer clock values cause leakage and fidelity loss.
Key takeaway. Precision of phase estimation limits HHL's accuracy.

Q.76 Single-correct MCQ · Memory with 8 clock qubits

With `NUM_CLOCK_QUBITS = 8` the default circuit has 11 qubits. The fp64 state vector needs ... bytes, and the notebook's memory line (state + one host copy) shows about ...

(A) 32 768; 32.8 kB **(B)** ✓ 32 768; 65.5 kB **(C)** 4 194 304; 8.4 MB **(D)** 2 048; 4.1 kB

Answer: (B) Detailed solution

Correct answer. (B) 32 768; 65.5 kB
Concepts required. Memory for 11 qubits.
Relevant formulae. $16 \times 2^{11} = 32\,768$ B, plus one copy.
Step-by-step explanation. $16 \times 2^{11} = 32\,768$ B; with the extra copy for `get_state`: 65 536 B $\approx$ 65.5 kB. Memory is never the limit for these small HHL instances; gate count and depth are (255 multi-controlled rotations at $n_c = 8$). $4\,194\,304 = 16 \times 2^{18}$ would be an 18-qubit state.
Why the other options are incorrect.
(A) The notebook adds the host copy: 65.5 kB.
(C) 4 194 304 B is an 18-qubit state.
(D) 2 048 B is the 7-qubit state.
Key takeaway. Depth, not memory, limits small HHL simulations.

Q.77 Single-correct MCQ · How e^{iAt} is implemented

Which statement correctly describes how `HHL.ipynb` implements $U = e^{iAt}$?

(A) It uses qubitisation with a block encoding of A
(B) It uses first-order Trotterisation of A's Pauli decomposition
(C) ✓ It computes e^{iAt} exactly with `scipy.linalg.expm` and applies each power U^{2^j} as a custom controlled gate; this is exact on simulators but is not a scalable Hamiltonian simulation
(D) It approximates e^{iAt} by $I + iAt$

Answer: (C) Detailed solution

Correct answer. (C) It computes e^{iAt} exactly with `scipy.linalg.expm` and applies each power U^{2^j} as a custom controlled gate; this is exact on simulators but is not a scalable Hamiltonian simulation
Concepts required. How e^{iAt} is implemented.
Relevant formulae. `expm(1j * A * t)`, `matrix_power`, `register_operation`.
Step-by-step explanation. `U_step = expm(1j * A_work * t_evolution)`; powers via `np.linalg.matrix_power` and `cudaq.register_operation`. A $2^{n_b} \times 2^{n_b}$ matrix per gate is fine for 2 system qubits, but costs exponential classical effort and has no gate decomposition. Real implementations need sparse Hamiltonian simulation (product formulas, qubitisation), whose cost enters HHL's s and κ dependence. $I + iAt$ is not even unitary.
Why the other options are incorrect.
(A) No block encoding or qubitisation in the notebook.
(B) No Trotterisation in the notebook.
(D) $I + iAt$ is not unitary.
Key takeaway. Exact custom gates are a simulator shortcut, not Hamiltonian simulation.

Q.78 Single-correct MCQ · Useful outputs

Which use of HHL can, in principle, keep an exponential advantage?

(A) replacing Gaussian elimination for 4×4 systems

(B) printing every component of x for a dense $10^6 \times 10^6$ matrix

(C) solving a dense, ill-conditioned system with $\kappa \sim d$

(D) ✓ estimating a quantity such as $x^\dagger M x$ for a sparse, well-conditioned A with efficiently preparable $|b\rangle$

Answer: (D) Detailed solution

Correct answer. **(D)** estimating a quantity such as $x^\dagger M x$ for a sparse, well-conditioned A with efficiently preparable $|b\rangle$

Concepts required. Where HHL can help.

Relevant formulae. estimate $x^\dagger M x$ with $O(1/\varepsilon^2)$ shots.

Step-by-step explanation. An expectation value needs only $O(1/\varepsilon^2)$ shots of $|x\rangle$, independent of d, so the $\log d$ scaling survives if s, κ and state preparation are cheap. Printing x costs d/ε^2 shots; $\kappa \sim d$ destroys the advantage through the κ dependence; small systems are trivially solved classically (the notebook's 4×4 example is a demonstration, not a speed-up).

Why the other options are incorrect.

(A) 4×4 systems are trivial classically.

(B) Printing all components costs d/ε^2 shots.

(C) $\kappa \sim d$ removes the advantage.

Key takeaway. Ask for a property of x, not x itself.

Paragraph for Questions 79 and 80

Previously recorded default run of `HHL.ipynb` (CUDA-Q 0.16.0, target `qpp-cpu`): 4×4 system, $\kappa = 4$, 7 qubits, $\|b\| = 1$, $C = 1$. Success probability: 0.355903 from the state, 0.3543 from 20 000 shots (7086 accepted). Exact $|x_i|^2/\|x\|^2 = (0.7622, 0.0598, 0.1476, 0.0305)$; from the accepted shots $(0.7691, 0.0587, 0.1420, 0.0302)$. Fidelity of the state read-out 1.0000000000, relative error 1.655×10^{-15}.

Q.79 Paragraph based · Recorded run: shot noise

The sampled $|x_0|^2/\|x\|^2 = 0.7691$ differs from the exact 0.7622 by 0.0069. With 7086 accepted shots, its binomial standard error is about ..., so the deviation is ...

(A) 0.0051; about 14 standard errors — a bug

(B) ✓ 0.0051; about 1.4 standard errors — consistent with shot noise

(C) 0.0034; about 2 standard errors

(D) 0.0119; about 0.6 standard errors

Answer: (B) Detailed solution

Correct answer. **(B)** 0.0051; about 1.4 standard errors — consistent with shot noise

Concepts required. Shot noise of a sampled probability.

Relevant formulae. $\sqrt{p(1-p)/S}$.

Step-by-step explanation. $\sqrt{p(1-p)/S} = \sqrt{0.7622 \times 0.2378/7086} = \sqrt{2.558 \times 10^{-5}} = 0.0051$; $0.0069/0.0051 \approx 1.4$ — an ordinary fluctuation. The notebook's quoted scale $1/\sqrt{7086} = 0.0119$ is a conservative upper bound (even the worst case $\sqrt{p(1-p)} = \frac{1}{2}$ gives only 0.0059). 0.0034 is the standard error of the overall success fraction from 20 000 shots.

Why the other options are incorrect.

(A) 14 standard errors mistakes 0.0069/0.0005.

(C) 0.0034 is the success-fraction error from 20000 shots.

(D) 0.0119 is the notebook's conservative scale.

Key takeaway. Compare deviations with the right standard error.

Q.80 PARAGRAPH BASED · Recorded run: $\|x\|$ from P_{success}

Using $P_{\text{success}} = C^2\|x\|^2/\|b\|^2$, the norm $\|x\|$ estimated from the shot-based success fraction is about

(A) 1.68 **(B)** 0.354 **(C)** ✓ 0.595 **(D)** 0.125

Answer: (C) Detailed solution

Correct answer. (C) 0.595

Concepts required. Norm from the success probability.

Relevant formulae. $\|x\| = \frac{\|b\|}{C}\sqrt{P_{\text{success}}}$.

Step-by-step explanation. $\|x\| = \frac{\|b\|}{C}\sqrt{P} = \sqrt{0.3543} = 0.595$ (exact $\sqrt{0.355903} = 0.5966$). So even on hardware the norm of x can be estimated from the success rate, while the components need the histogram (magnitudes) plus extra measurements (signs). 0.354 forgets the square root; $1.68 = 1/0.595$.

Why the other options are incorrect.

(A) 1.68 is the reciprocal.

(B) 0.354 forgets the square root.

(D) 0.125 has no derivation.

Key takeaway. Even on hardware, $\|x\|$ follows from the success rate.